

Mathematics A

Unit 1 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

68 classified items • 259 marks • 19 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Unit 1 • Expertise • Unit 1 • Expertise Q16+ • 68 items • 259 marks

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June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Set Notation & Venn Diagrams

Unit 1 • Expertise • 7 marks • 3 item(s)

January 2021 | 4MA1-2H Q16 | 3 marks

Complete or solve a practical three-set survey

January 2021 | 4MA1-2H Q16 | 3 marks

Complete or solve a practical three-set survey

June 2021 | 4MA1-1H Q19 | 1 marks

Read and combine three-set regions

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Fractions, Decimals & Percentages

Unit 1 • Expertise • 7 marks • 3 item(s)

June 2025 | 4MA1-2H Q18 | 2 marks

Pure recurring block

November 2021 | 4MA1-1H Q18 | 3 marks

Non-recurring prefix followed by a recurring block

January 2023 | 2H Q16 | 2 marks

Decimal to fraction or percentage

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Rounding, Estimation & Bounds

Unit 1 • Expertise • 16 marks • 4 item(s)

June 2025 | 4MA1-2H Q22 | 4 marks

Geometry, rate or practical decision bounds

November 2024 | 4MA1-1H Q21 | 3 marks

Algebraic expression upper or lower bound

May-Nov 2020 | 1H Q17 | 4 marks

Mixed stated accuracies

January 2023 | 1H Q19 | 5 marks

Round to decimal places or significant figures

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Surds

Unit 1 • Expertise • 14 marks • 5 item(s)

June 2023 | 4MA1-2H Q23 | 4 marks

Binomial surd denominator using a conjugate

November 2024 | 4MA1-1H Q17 | 1 marks

Extract square factors from a surd

June 2022 | 4MA1-1H Q16 | 3 marks

Add or subtract like surds

January 2020 | 4MA1-1H Q17 | 3 marks

Expand binomials containing surds

May-Nov 2020 | 1H Q19 | 3 marks

Single-term surd denominator

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Algebraic Roots & Indices

Unit 1 • Expertise • 11 marks • 3 item(s)

June 2024 | 4MA1-2H Q23 | 3 marks

Complex indexed monomial or fractional outer power

June 2022 | 4MA1-2H Q24 | 3 marks

Express one exponent variable in terms of others

January 2023 | 2H Q26 | 5 marks

Zero and negative indices

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Expanding Brackets

Unit 1 • Expertise • 3 marks • 1 item(s)

November 2023 | 4MA1-2H Q16 | 3 marks

Three linear brackets

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Factorising

Unit 1 • Expertise • 2 marks • 1 item(s)

November 2021 | 4MA1-1H Q17 | 2 marks

Non-monic quadratic ax^2+bx+c

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Completing the Square

Unit 1 • Expertise • 7 marks • 2 item(s)

June 2023 | 4MA1-2H Q19 | 3 marks

Positive non-unit leading coefficient

January 2022 | 4MA1-1H Q20 | 4 marks

Negative leading coefficient or maximum form

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Algebraic Fractions

Unit 1 • Expertise • 18 marks • 4 item(s)

November 2023 | 4MA1-2H Q24 | 5 marks

Equation containing algebraic fractions

November 2024 | 4MA1-2H Q23 | 4 marks

Divide factorisable algebraic fractions

January 2021 | 4MA1-2H Q21 | 4 marks

Multiply factorisable algebraic fractions

May-Nov 2020 | 1H Q21 | 5 marks

Substitute a variable relation into a fraction

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Coordinate Geometry

Unit 1 • Expertise • 23 marks • 4 item(s)

June 2022 | 4MA1-2H Q22 | 7 marks

Kite or symmetric vertex coordinates

January 2020 | 4MA1-1H Q22 | 6 marks

Coordinates from isosceles and gradient constraints

January 2021 | 4MA1-2H Q18 | 4 marks

Rectangle or parallelogram missing vertex

January 2023 | 1H Q23 | 6 marks

Read a plotted coordinate pair

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Linear Graphs ($y = mx + c$)

Unit 1 • Expertise • 10 marks • 3 item(s)

June 2024 | 4MA1-2H Q18 | 2 marks

Parallel and perpendicular gradient relation

January 2020 | 4MA1-2H Q19 | 1 marks

Parallel line through a point

November 2021 | 4MA1-1H Q20 | 7 marks

Perpendicular line through a point or midpoint

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Graphs of Functions

Unit 1 • Expertise • 6 marks • 2 item(s)

June 2025 | 4MA1-2H Q25 | 3 marks

Feature-led polynomial sketch

June 2023 | 4MA1-2H Q18 | 3 marks

Recognise reciprocal or rational graph shape

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Area & Perimeter

Unit 1 • Expertise • 7 marks • 2 item(s)

January 2021 | 4MA1-2H Q18 | 3 marks

Find perimeter of a simple triangle or rectangle

January 2021 | 4MA1-2H Q20 | 4 marks

Find composite or shaded polygonal area

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Circles, Arcs & Sectors

Unit 1 • Expertise • 28 marks • 6 item(s)

November 2023 | 4MA1-2H Q20 | 4 marks

Find annular, overlapping or multi-circle area

June 2023 | 4MA1-1H Q22 | 6 marks

Solve an algebraic circle-measure constraint

November 2024 | 4MA1-1H Q16 | 4 marks

Find sector area

November 2024 | 4MA1-2H Q25 | 5 marks

Find annular, overlapping or multi-circle area

January 2022 | 4MA1-1H Q16 | 4 marks

Recover radius or central angle from an arc/sector measure

January 2023 | 1H Q21 | 5 marks

Find a circular segment area or perimeter

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Sine, Cosine Rule & Area of Triangles

Unit 1 • Expertise • 26 marks • 5 item(s)

June 2025 | 4MA1-1H Q20 | 5 marks

Use two non-right triangles sharing a diagonal

June 2023 | 4MA1-2H Q22 | 5 marks

Combine non-right trigonometry with circle or elevation geometry

June 2022 | 4MA1-1H Q18 | 5 marks

Find an acute angle using the sine rule

January 2022 | 4MA1-2H Q23 | 6 marks

Resolve the ambiguous sine-rule angle

January 2023 | 1H Q16 | 5 marks

Find a side using the sine rule

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

3D Pythagoras & Trigonometry

Unit 1 • Expertise • 19 marks • 4 item(s)

January 2022 | 4MA1-1H Q21 | 5 marks

Use midpoint or symmetry structure in a 3D model

June 2021 | 4MA1-1H Q22 | 4 marks

Use a non-rectangular base before a line-plane angle

May-Nov 2020 | 2H Q18 | 5 marks

Find angle between a line and a plane

January 2023 | 2H Q23 | 5 marks

Use 3D trigonometry to obtain a terminal volume

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Histograms

Unit 1 • Expertise • 8 marks • 4 item(s)

June 2022 | 4MA1-1H Q21 | 3 marks

Estimate a partial-class frequency

June 2021 | 4MA1-1H Q18 | 2 marks

Complete a table from a partial histogram

June 2021 | 4MA1-1H Q18 | 1 marks

Complete missing histogram bars from mixed information

June 2021 | 4MA1-1H Q18 | 2 marks

Estimate a histogram proportion, fraction or percentage

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Probability Diagrams - Venn & Tree Diagrams

Unit 1 • Expertise • 11 marks • 3 item(s)

June 2024 | 4MA1-2H Q16 | 7 marks

Read regions and probabilities from a Venn diagram

November 2021 | 4MA1-1H Q16 | 2 marks

Solve unknown Venn regions using a total

November 2021 | 4MA1-1H Q16 | 2 marks

Complete a three-set Venn diagram

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Combined & Conditional Probability

Unit 1 • Expertise • 36 marks • 9 item(s)

June 2025 | 4MA1-1H Q21 | 3 marks

Calculate an exact composition in three or more draws

June 2025 | 4MA1-2H Q20 | 6 marks

Calculate a direct given-that probability

June 2024 | 4MA1-1H Q16 | 4 marks

Use a complement in repeated no-replacement selection

November 2023 | 4MA1-1H Q20 | 4 marks

Find exactly one, same or different independent outcomes

June 2022 | 4MA1-2H Q16 | 8 marks

Calculate a direct given-that probability

January 2020 | 4MA1-2H Q18 | 2 marks

Use a histogram or table as a conditional sample space

November 2021 | 4MA1-1H Q16 | 1 marks

Calculate a direct given-that probability

November 2021 | 4MA1-2H Q16 | 3 marks

Compare with-replacement and without-replacement models

May-Nov 2020 | 2H Q20 | 5 marks

Recover an unknown population from no-replacement probability

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

16 Some students were asked the following question.

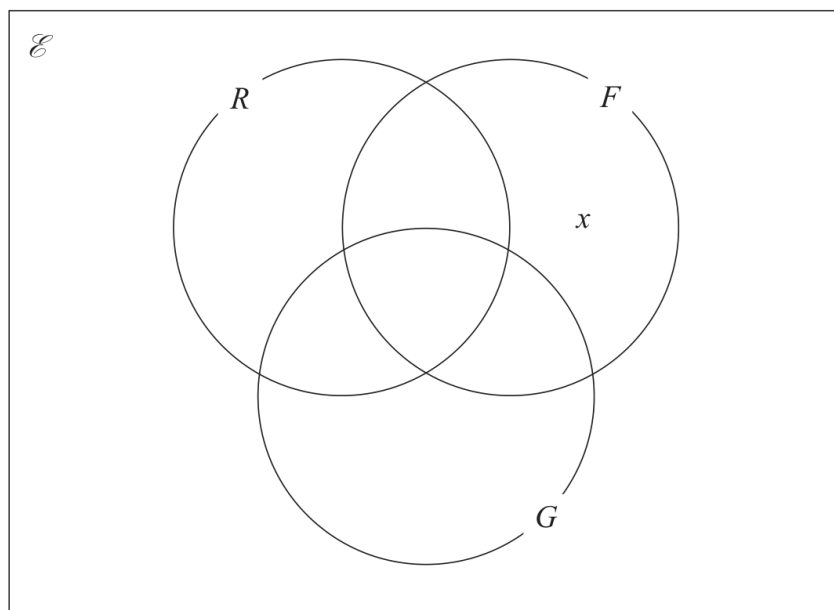
“Which of the subjects Russian (R), French (F) and German (G) do you study?”

Of these students

- 4 study all three of Russian, French and German
- 10 study Russian and French
- 13 study French and German
- 6 study Russian and German
- 24 study German
- 11 study none of the three subjects
- the number who study Russian only is twice the number who study French only.

Let x be the number of students who study French only.

(a) Show all this information on the Venn diagram, giving the number of students in each appropriate subset, in terms of x where necessary.



(3)

Given that the number of students who were asked the question was 80

(b) work out the number of these students that study Russian.

(3)

Worked solution

January 2021 | Question 16 | 3 marks

Family: Resolve a three-set Venn model • Variant: Complete or solve a practical three-set survey

Method / working

1. Place 4 in the triple intersection, $10-4=6$ in R intersection F only, $13-4=9$ in F intersection G only, and $6-4=2$ in R intersection G only.

2. Let French-only= x and Russian-only= $2x$; German-only= $24-4-9-2=9$.

3. Place 11 outside and label every region on the supplied Venn diagram.

Final answer: Completed Venn diagram with x , $2x$, 6, 2, 4, 9, 9, 11

16 Some students were asked the following question.

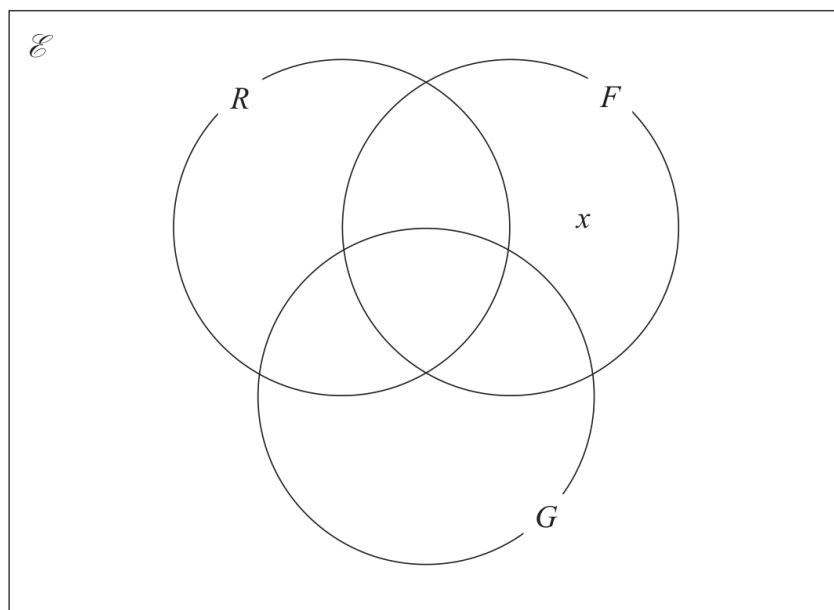
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Given that the number of students who were asked the question was 80

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(3)

Worked solution

January 2021 | Question 16 | 3 marks

Family: Resolve a three-set Venn model • Variant: Complete or solve a practical three-set survey

Method / working

1. Use the total: $2x+6+x+2+4+9+9+11=80$.

2. Solve $3x+41=80$ to get $x=13$.

3. Russian total= $2x+6+2+4=38$.

Final answer: 38 students

(iii) $n([A \cap B] \cup C)$

(1)

Worked solution

June 2021 | Question 19 | 1 marks

Family: Resolve a three-set Venn model • Variant: Read and combine three-set regions

Method / working

1. Count $(A \cap B) \cup C$ from the labelled regions: 28.

Final answer: 28

4MA1-2H Q18

2 marks

18 Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$

(Total for Question 18 is 2 marks)

WORKING SPACE

4MA1-2H Q18

2 marks

18 Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$

(Total for Question 18 is 2 marks)

Convert a recurring decimal to a fraction**2 marks****Align a complete repeating block**

$$x = 0.\overline{702}, \quad 1000x = 702.\overline{702}.$$

Subtract and simplify

$$999x = 702 \Rightarrow x = \frac{702}{999} = \frac{26}{37}.$$

Final answer

$$\frac{26}{37}$$

- 18** $0.4\dot{x}$ is a recurring decimal.
 x is a whole number such that $1 \leq x \leq 9$

Find, in terms of x , the recurring decimal $0.4\dot{x}$ as a fraction.
Give your fraction in its simplest form.
Show clear algebraic working.

Worked solution

November 2021 | Question 18 | 3 marks

Family: Convert a recurring decimal to a fraction • Variant: Non-recurring prefix followed by a recurring block

Method / working

1. Let

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $y = 0.4\dot{x}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. So

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $y = 0.4xxx\ldots$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. Multiply by (10) and (100) :

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $10y = 4.xxx\ldots$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $100y = 4x.xxx\ldots$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. Subtract:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. $100y - 10y = (4x.xxx\ldots) - (4.xxx\ldots)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $90y = x + 36$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. Therefore

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $y = \frac{x+36}{90}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. Answer: $\frac{x+36}{90}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $\frac{x+36}{90}$

16 Use algebra to show that $0.\dot{4}\dot{3}\dot{8} = \frac{217}{495}$

(Total for Question 16 is 2 marks)

Answer reference | January 2023 | Question 16

Family: Convert terminating representations • Variant: Decimal to fraction or percentage

Question	Working	Answer	Mark	Notes
16	eg $1000x = 438.38\ldots$ $10x = 4.38\ldots$ or $100x = 43.838\ldots$ $x = 0.438\ldots$ oe		2	M1 For selecting 2 correct recurring decimals that when subtracted give a whole number or terminating decimal (43.4 or 434 etc) eg $1000x = 438.38\ldots$ and $10x = 4.38\ldots$ or $100x = 43.838\ldots$ and $x = 0.438\ldots$ with intention to subtract. (if recurring dots not shown then showing at least one of the numbers to at least 5sf) or $0.4 + 0.\dot{0}3\dot{8}$ and eg $1000x = 38.38\ldots$ & $10x = 0.3838\ldots$, with intention to subtract.
	eg $1000x - 10x = 438.38\ldots - 4.38\ldots = 434$ and $\frac{434}{990} = \frac{217}{495}$ or eg $100x - x = 43.838\ldots - 0.438\ldots = 43.4$ and $\frac{43.4}{99} = \frac{217}{495}$ or eg $1000x - 10x = 38.38\ldots - 0.3838 = 38$ and $0.4 + \frac{38}{990} = \frac{4 \times 99 + 38}{990} = \frac{434}{990} = \frac{217}{495}$ oe <i>working required</i>	Clearly shown		A1 For completion to $\frac{217}{495}$ dep on M1 and use of some algebra
Total 2 marks				

Worked solution

January 2023 | Question 16 | 2 marks

Family: Convert terminating representations • Variant: Decimal to fraction or percentage

Official final mark-scheme pages are embedded immediately after this question.

4MA1-2H Q22

4 marks

22 The diagram shows a square inside rectangle $ABCD$

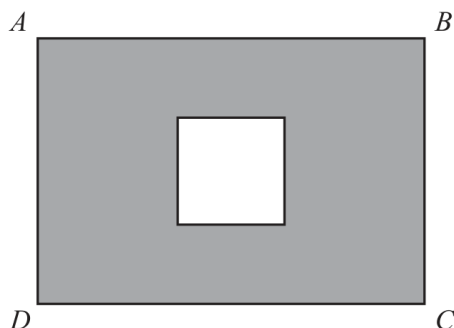


Diagram **NOT**
accurately drawn

The total area of the region shown shaded in the diagram is $X \text{ cm}^2$

$AB = 11.5 \text{ cm}$ correct to the nearest 0.5 cm

$BC = 9.2 \text{ cm}$ correct to 2 significant figures

side of square = 4.1 cm correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 22 is 4 marks)

WORKING SPACE

4MA1-2H Q22

4 marks

22 The diagram shows a square inside rectangle $ABCD$

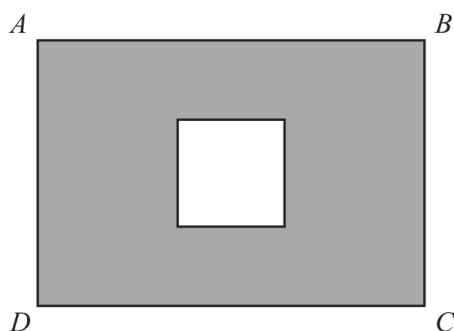


Diagram **NOT**
accurately drawn

The total area of the region shown shaded in the diagram is $X \text{ cm}^2$

$AB = 11.5 \text{ cm}$ correct to the nearest 0.5 cm

$BC = 9.2 \text{ cm}$ correct to 2 significant figures

side of square $= 4.1 \text{ cm}$ correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 22 is 4 marks)

Bounds for a shaded area**4 marks****Input bounds**

$$11.25 \leq AB < 11.75, \quad 9.15 \leq BC < 9.25, \quad 4.05 \leq s < 4.15.$$

Upper shaded area

$$X_U = 11.75(9.25) - 4.05^2 = 92.285.$$

Lower shaded area

$$X_L = 11.25(9.15) - 4.15^2 = 85.715.$$

Choose the safe accuracy

$$85.715 \leq X < 92.285; \text{ both bounds round to 90 to 1 significant figure.}$$

Final answer

$$X = 90 \text{ cm}^2 \text{ (1 s.f.)}$$

4MA1-1H Q21 M01

3 marks

$$21 \quad T = \frac{x^2 + y^2}{w}$$

 $x = 28.4$ correct to 1 decimal place. $y = 17$ correct to 2 significant figures. $w = 90$ correct to the nearest 5Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.

Show your working clearly.

.....

WORKING SPACE

Use upper and lower bounds

3 marks

M01 – Use upper and lower bounds**Write the bounds**

$$UB(x) = 28.45, UB(y) = 17.5, LB(w) = 87.5.$$

Substitute

$$UB(T) = \frac{28.45^2 + 17.5^2}{87.5} = 12.7503\dots$$

Round

$$UB(T) = 12.8 \text{ (3 s.f.)}.$$

Final answer

12.8

- 17** A metal block has a mass of 5 kg, correct to the nearest 50 grams.
The block has a volume of $(1.84 \times 10^{-3}) \text{ m}^3$, correct to 3 significant figures.

Work out the upper bound for the density of the block.
Give your answer in kg/m^3 correct to 1 decimal place.
Show your working clearly.

..... kg/m^3

(Total for Question 17 is 4 marks)

Answer reference | May-Nov 2020 | Question 17

Family: Recover single-measurement bounds • Variant: Mixed stated accuracies

17	5025 or 5.025 or 4975 or 4.975		4	B1 Accept 5024.9 for 5025 or 5.0249 for 5.025
	1.845×10^{-3} oe or 1.835×10^{-3} oe			B1 Accept 1.8449×10^{-3} for 1.845×10^{-3}
	$\frac{5.025}{1.835 \times 10^{-3}} (= 2738.4...) \text{ oe}$			M1 for correct substitution into $\frac{m_{UB}}{v_{LB}}$ where $5 < m_{UB} \leq 5.025$ and $1.835 \times 10^{-3} \leq v_{LB} < 1.84 \times 10^{-3}$
		2738.4		A1 dep on correct working
				Total 4 marks

Worked solution

May-Nov 2020 | Question 17 | 4 marks

Family: Recover single-measurement bounds • Variant: Mixed stated accuracies

Method

Density is

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

To get the upper bound for density, use the upper bound for mass and the lower bound for volume.

The mass is 5 kg correct to the nearest 50 g.

$$50 \text{ g} = 0.05 \text{ kg}$$

Half of this is 0.025 kg, so

$$\text{upper bound for mass} = 5 + 0.025 = 5.025$$

The volume is $1.84 \times 10^{-3} = 0.00184 \text{ m}^3$, correct to 3 significant figures.

The lower bound is

$$0.00184 - 0.000005 = 0.001835$$

So the upper bound for the density is

$$(5.025) / (0.001835) = 2738.419 \dots$$

Correct to 1 decimal place,

$$2738.4$$

Answer: 2738.4 kg/m^3

19 The acceleration, a , of an object is given by

$$a = \frac{v - u}{t}$$

where

$v = 45.23$ correct to 2 decimal places

$u = 5.12$ correct to 2 decimal places

$t = 8.5$ correct to 2 significant figures

By considering bounds, work out the value of a to a suitable degree of accuracy.
Show your working clearly and give a reason for your answer.

$a = \dots\dots\dots$

(Total for Question 19 is 5 marks)

Answer reference | January 2023 | Question 19

Family: Round to a stated accuracy • Variant: Round to decimal places or significant figures

19	45.225 or 45.235 or 5.115 or 5.125 or 8.45 or 8.55		5	B2 for all 6 correct (B1 for 4 or 5 correct) Accept $45.234\dot{9}$ for 45.235 $5.124\dot{9}$ for 5.125 $8.54\dot{9}$ for 8.55
	$\frac{45.235 - 5.115}{8.45} (= 4.7479\dots)$			M1 for correct substitution into the <i>UB</i> $a = \frac{v-u}{t}$ where $45.23 < v_{(UB)} \leq 45.235$ $5.115 \leq u_{(LB)} < 5.12$ $8.45 \leq t_{(LB)} < 8.5$
	$\frac{45.225 - 5.125}{8.55} (= 4.6900\dots)$			M1 for correct substitution into the <i>LB</i> $a = \frac{v-u}{t}$ where $45.225 \leq v_{(LB)} < 45.23$ $5.12 < u_{(UB)} \leq 5.125$ $8.5 < t_{(UB)} \leq 8.55$
	Working required	4.7 and correct reason		A1 dep on M2 4.7 and both answers round to 4.7 oe e.g. 1 dp or 2 sf
Total 5 marks				

Worked solution

January 2023 | Question 19 | 5 marks

Family: Round to a stated accuracy • Variant: Round to decimal places or significant figures

Official final mark-scheme pages are embedded immediately after this question.

4MA1-2H Q23

4 marks

23 The diagram shows a cuboid with a square cross section.

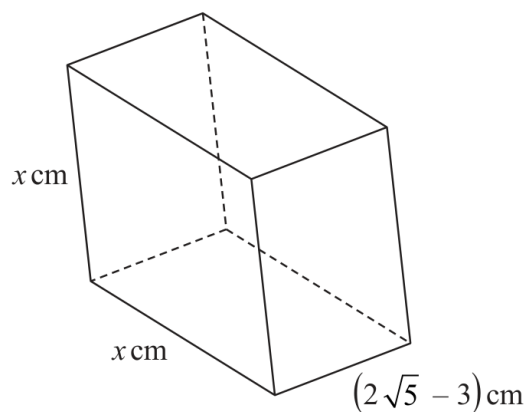


Diagram **NOT**
accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\text{cm}^3$

Without using a calculator, find the value of x

Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

$x = \dots\dots\dots$

Continue your method**4 marks**

WORKING SPACE

4MA1-2H Q23

4 marks

- 23 The diagram shows a cuboid with a square cross section.

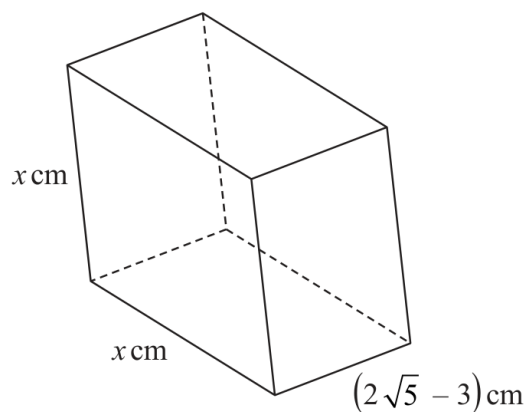


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Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

$x = \dots\dots\dots$

Surd volume and a square cross-section**4 marks****Form x squared**

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5} \Rightarrow x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}.$$

Rationalise

$$x^2 = \frac{(13 + 6\sqrt{5})(2\sqrt{5} + 3)}{11} = 9 + 4\sqrt{5}.$$

Recognise the square

$$9 + 4\sqrt{5} = (2 + \sqrt{5})^2.$$

Take the positive length

$$x = 2 + \sqrt{5}.$$

Final answer

$$2 + \sqrt{5} \text{ cm}$$

4MA1-1H Q17 M01

1 marks

17 (a) Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

.....
(1)

WORKING SPACE

Express a surd in the required form

1 marks

M01 – Express a surd in the required form**Factor inside the root**

$$\sqrt{675} = \sqrt{25 \times 27} = 5\sqrt{27}.$$

Final answer

$$n = 5$$

4MA1-1H Q16 Q16

3 marks

- 16 Without using a calculator, show that $\frac{12}{\sqrt{2}-1} - (\sqrt{2})^5 = 2\sqrt{32} + 12$
Show your working clearly.

WORKING SPACE

4MA1-1H Q16 Q16

3 marks

- 16 Without using a calculator, show that $\frac{12}{\sqrt{2}-1} - (\sqrt{2})^5 = 2\sqrt{32} + 12$
Show your working clearly.

Chapter: Surds

3 marks

Q16 – Chapter: Surds

Family: Simplify and collect surds • Variant: Add or subtract like surds

Method

$$12/(\sqrt{2} - 1) = 12(\sqrt{2} + 1)/[(\sqrt{2} - 1)(\sqrt{2} + 1)] = 12\sqrt{2} + 12.$$

Why: Multiply by the conjugate to rationalise the denominator.

Method

$$(\sqrt{2})^5 = 4\sqrt{2}, \text{ so the left side is } 12\sqrt{2} + 12 - 4\sqrt{2} = 8\sqrt{2} + 12.$$

Why: Use powers of $\sqrt{2}$ and collect like surds.

Method

Since $\sqrt{32} = 4\sqrt{2}$, the right side $2\sqrt{32} + 12$ is also $8\sqrt{2} + 12$; therefore the identity is shown.

Why: Simplify both sides to the same exact surd form.

Final answer

shown identity: Both sides simplify to $8\sqrt{2} + 12$

17 (a) Show that $(6 + 2\sqrt{12})^2 = 12(7 + 4\sqrt{3})$

Show each stage of your working.

(3)

Worked solution

January 2020 | Question 17 | 3 marks

Family: Multiply and expand surd expressions • Variant: Expand binomials containing surds

Method / working

1. $\sqrt{12} = 2\sqrt{3}$, so $6 + 2\sqrt{12} = 6 + 4\sqrt{3}$.

2. Square: $(6 + 4\sqrt{3})^2 = 84 + 48\sqrt{3}$.

3. Factor 12: $84 + 48\sqrt{3} = 12(7 + 4\sqrt{3})$.

Final answer: $12(7 + 4\sqrt{3})$

19 Without using a calculator, rationalise the denominator of $\frac{6}{3 - \sqrt{7}}$

Simplify your answer.

You must show each stage of your working.

(Total for Question 19 is 3 marks)

Answer reference | May-Nov 2020 | Question 19

Family: Rationalise a surd denominator • Variant: Single-term surd denominator

19	$\frac{6}{3-\sqrt{7}} \times \frac{3+\sqrt{7}}{3+\sqrt{7}}$ or $\frac{6}{3-\sqrt{7}} \times \frac{-3-\sqrt{7}}{-3-\sqrt{7}}$			M1
	$\frac{6(3+\sqrt{7})}{3^2-7}$ or $\frac{6(3+\sqrt{7})}{2}$ or $\frac{6(-3-\sqrt{7})}{-3^2+7}$ or $\frac{6(-3-\sqrt{7})}{-2}$			M1 (numerator may be expanded or denominator may be 4 terms which need to be all correct)
		$9+3\sqrt{7}$	3	A1 dep on M2 for $9+3\sqrt{7}$ or $3(3+\sqrt{7})$ from correct working
				Total 3 marks

Worked solution

May-Nov 2020 | Question 19 | 3 marks

Family: Rationalise a surd denominator • Variant: Single-term surd denominator

Method

$$(6)/(3-\sqrt{7})$$

Rationalise the denominator:

$$(6)/(3-\sqrt{7}) \times (3+\sqrt{7})/(3+\sqrt{7})$$

$$=(6(3+\sqrt{7}))/((9-7))$$

$$=(6(3+\sqrt{7}))/2=3(3+\sqrt{7})$$

$$=9+3\sqrt{7}$$

Answer: $9+3\sqrt{7}$

4MA1-2H Q23

3 marks

23 Simplify $\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

WORKING SPACE

4MA1-2H Q23

3 marks

23 Simplify $\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

Surds and index laws

3 marks

Rewrite in base 5

$$30 = 6 \cdot 5, 25^{2x+7} = 5^{4x+14}, \sqrt{180} = 6\sqrt{5}.$$

Combine

$$\frac{30 \cdot 25^{2x+7}}{\sqrt{180}(\sqrt{5})^{4x+9}} = 5^{2x+10}.$$

Final answer

$$5^{2x+10}$$

4MA1-2H Q24 Q24

3 marks

24

$$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

WORKING SPACE

4MA1-2H Q24 Q24

3 marks

24

$$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

Chapter: Algebraic Roots & Indices

3 marks

Q24 – Chapter: Algebraic Roots & Indices

Family: Solve an index relation by a common base • Variant: Express one exponent variable in terms of others

Method

Use $18 = 2 \times 3^2$, $\sqrt{27} = 3^{(3/2)}$, $6 = 2 \times 3$, and $9 = 3^2$.

Why: Rewrite every factor using base 3 where possible.

Method

The left side becomes $[2 \times 3^2 \times 3^{(6n+9)}] / [2 \times 3 \times 3^{(4n+16)}]$.

Why: Multiply the nested exponents and collect the fixed factors.

Method

This simplifies to $3^{(6n+11-4n-17)} = 3^{(2n-6)}$, so $x = 2n - 6$.

Why: Subtract denominator exponents and equate equal bases.

Final answer

x: $x = 2n - 6$

26 Find the values of n such that

$$\frac{10^{4n} \times 2^{3(n^2-5n)} \times 5^{2(1-2n)}}{20^2} = 1$$

Show clear algebraic working.

(Total for Question 26 is 5 marks)

Answer reference | January 2023 | Question 26

Family: Interpret zero, negative and fractional indices • Variant: Zero and negative indices

Q	working	Answer	Mark	Notes
26	For one of: $10^{4n} = (5 \times 2)^{4n}$ or $5^{4n} \times 2^{4n}$ or oe $20^2 = (5 \times 2^2)^2$ or $5^2 \times 2^4$ or $5^2 \times 2^2 \times 2^2$ or cancelling 5^2 on numerator with 20^2 to get 16 or 2^4		5	M1 for writing 10^{4n} correctly as a product of 5 and 2 to the power $4n$ oe or writing 20^2 correctly as a product of 5 and 2^2 to the power 2 oe or cancelling 5^2 on numerator with 20^2 to get 16 or 2^4
	for getting numerator to the stage (this scores M1M1) $2^{4n} \times 2^{3n^2-15n} \times 5^2$ oe eg $2^{3n^2-11n} \times 5^{-4n+4n+2}$ oe or for two of $10^{4n} = (5 \times 2)^{4n}$ or $5^{4n} \times 2^{4n}$ or oe $20^2 = (5 \times 2^2)^2$ or $5^2 \times 2^4$ or $5^2 \times 2^2 \times 2^2$ or cancelling 5^2 on numerator with 20^2 to get 16 or 2^4 or getting the numerator to the stage			M1
	$\frac{2^{4n} \times 2^{3n^2} \times 2^{-15n}}{2^4} [= 1 \text{ or } 2^0]$ or $2^{4n} \times 2^{3n^2} \times 2^{-15n} = 2^4$ oe			M1 For writing the equation in powers of 2 only
	$3n^2 - 11n - 4 [= 0]$			A1 Correct quadratic equation dep on M1
	working required	$-\frac{1}{3}, 4$		A1 Both answers required dep on M1
Total 5 marks				

Worked solution

January 2023 | Question 26 | 5 marks

Family: Interpret zero, negative and fractional indices • Variant: Zero and negative indices

Official final mark-scheme pages are embedded immediately after this question.

4MA1-2H Q16 whole

3 marks

16 Expand and simplify $(2x + 3)(x - 5)(x + 4)$

.....

WORKING SPACE

4MA1-2H Q16 whole

3 marks

16 Expand and simplify $(2x + 3)(x - 5)(x + 4)$

.....

Expand three brackets

3 marks

whole – Expand three brackets

Multiply two brackets

$$(x - 5)(x + 4) = x^2 - x - 20.$$

Multiply by the third bracket

$$(2x + 3)(x^2 - x - 20) = 2x^3 - 2x^2 - 40x + 3x^2 - 3x - 60.$$

Collect like terms

$$2x^3 + x^2 - 43x - 60.$$

Final answer

$$2x^3 + x^2 - 43x - 60$$

17 (a) Factorise $6y^2 - y - 5$

(2)

Worked solution

November 2021 | Question 17 | 2 marks

Family: Factorise a non-monic or special quadratic • Variant: Non-monic quadratic ax^2+bx+c

Method / working

1. (a)

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $6y^2-y-5=(6y+5)(y-1)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $(6y+5)(y-1)$, $f=\frac{8w-3}{w+2}$, $4(x-1)^2+3$

4MA1-2H Q19

3 marks

19 Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

WORKING SPACE

4MA1-2H Q19

3 marks

19 Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

Complete the square**3 marks****Group the quadratic terms**

$$3x^2 - 6x + 5 = 3(x^2 - 2x) + 5.$$

Complete the square

$$3(x^2 - 2x) + 5 = 3[(x - 1)^2 - 1] + 5 = 3(x - 1)^2 + 2.$$

Final answer

$$3(x - 1)^2 + 2$$

4MA1-1H Q20 Q20(a), Q20(b)

4 marks

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

.....
(3)

C is the curve with equation $y = 7 + 12x - 3x^2$

The point *A* is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of *A*

(.....,)
(1)

WORKING SPACE

4MA1-1H Q20 Q20(a), Q20(b)

4 marks

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

.....
(3)

C is the curve with equation $y = 7 + 12x - 3x^2$
The point A is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of A

(.....,)
(1)

Chapter: Completing the Square

4 marks

Q20(a) – Chapter: Completing the Square

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

Method

$$7+12x-3x^2=7-3(x^2-4x).$$

Why: Factor -3 from the quadratic terms.

Method

$$x^2-4x=(x-2)^2-4.$$

Why: Complete the square.

Method

$$\text{Therefore } 7+12x-3x^2=19-3(x-2)^2.$$

Why: Substitute the completed square and simplify.

Final answer

completed square: $19-3(x-2)^2$

Q20(b) – Chapter: Completing the Square

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

Method

The maximum occurs when $(x-2)^2=0$, giving $A=(2,19)$.

Why: The negative coefficient means the squared term is subtracted from 19.

Final answer

maximum point: $(2,19)$

4MA1-2H Q24 whole

5 marks

24 Solve $\frac{45x^2 - 80x}{3x^2 + x - 4} \times \left(\frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x = \dots\dots\dots$

WORKING SPACE

4MA1-2H Q24 whole

5 marks

24 Solve $\frac{45x^2 - 80x}{3x^2 + x - 4} \times \left(\frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x =$

Solve a rational algebraic equation

5 marks

whole – Solve a rational algebraic equation

Combine the bracketed fractions

$$\frac{1}{3x-4} + \frac{1}{x} = \frac{x+3x-4}{x(3x-4)} = \frac{4(x-1)}{x(3x-4)}.$$

Factor the first fraction

$$45x^3 - 80x = 5x(3x-4)(3x+4), \quad 3x^2 + x - 4 = (3x+4)(x-1).$$

Cancel common factors

The left-hand side simplifies to 20 (with the original denominator restrictions retained).

Solve the remaining equation

$$20 = \frac{4(x+2)}{5x-8} \Rightarrow 20(5x-8) = 4(x+2).$$

Find and check x

$$96x = 168 \Rightarrow x = \frac{7}{4}, \text{ which is not an excluded denominator value.}$$

Final answer

$$x = 7/4$$

4MA1-2H Q23 M01

4 marks

23 Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.

Show clear algebraic working.

WORKING SPACE

Simplify an algebraic fraction**4 marks****M01 – Simplify an algebraic fraction****Factorise the first numerator**

$$16x^2 - 36 = 4(2x - 3)(2x + 3).$$

Factorise the remaining terms

$$2x^2 + 7x + 6 = (2x + 3)(x + 2), \quad x^2 - 5x - 14 = (x - 7)(x + 2).$$

Cancel

The quotient simplifies to $4(2x-3)$.

Subtract the linear expression

$$4(2x-3)-(7+8x)=-19, \text{ so } n=-19.$$

Final answer

$$n = -19$$

21 Write $\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

Worked solution

January 2021 | Question 21 | 4 marks

Family: Multiply or divide algebraic fractions • Variant: Multiply factorisable algebraic fractions

Method / working

1. Factor $25x^2-64=(5x-8)(5x+8)$, $x^2-8x+15=(x-5)(x-3)$, and $5x^2-13x-6=(5x+2)(x-3)$.
2. Factor and cancel common factors before combining.
3. Put the remaining terms over a common denominator.
4. The simplified single fraction is $54/(5x+2)$.

Final answer: $54/(5x+2)$

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

(Total for Question 21 is 5 marks)

Answer reference | May-Nov 2020 | Question 21

Family: Substitute or transform a rational expression • Variant: Substitute a variable relation into a fraction

21	$\left(\frac{9x^2 - 4}{3x^2 - 13x - 10} = \right) \frac{(3x+2)(3x-2)}{(3x+2)(x-5)}$			M1 for either $(3x+2)(3x-2)$ or $(3x+2)(x-5)$	M2 for $\frac{9x^2 - 4}{(9x^2 - 4)(x-5)} =$ $\frac{1}{(x-5)}$
	$\left(\frac{9x^2 - 4}{3x^2 - 13x - 10} = \right) \frac{(3x+2)(3x-2)}{(3x+2)(x-5)}$			M1 for $(3x+2)(3x-2)$ and $(3x+2)(x-5)$	
	E.g. of denominators $(3x-2)(3x^2-13x-10)(x-1)$ or $(3x-2)(3x+2)(x-5)(x-1)$ or $9x^4 - 54x^3 + 41x^2 + 24x - 20$ or $(3x+2)(x-5)(x-1)$ or $3x^3 - 16x^2 + 3x + 10$ or $(3x-2)(x-5)(x-1)$ or $3x^3 - 20x^2 + 27x - 10$ or $(x-5)(x-1)$ or $x^2 - 6x + 5$			M1 (indep) ft their fractions for use of a correct common denominator for 2 fractions with algebraic denominators NB: fractions need not be simplified	
	$\frac{x-1-7(x-5)}{(x-5)(x-1)}$ or $\frac{x-1-7x+35}{(x-5)(x-1)}$ or $\frac{x-1-7(x-5)}{x^2-6x+5}$ or $\frac{x-1-7x+35}{x^2-6x+5}$ oe			M1 for a correct fraction with a correct quadratic denominator – may or may not be expanded which leads to a correct answer	
		$\frac{2(17-3x)}{(x-5)(x-1)}$	5	A1 accept $\frac{34-6x}{(x-5)(x-1)}$ oe; if denominator is expanded then it must be correct	
Total 5 marks					

Worked solution

May-Nov 2020 | Question 21 | 5 marks

Family: Substitute or transform a rational expression • Variant: Substitute a variable relation into a fraction

Method

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

Factor:

$$9x^2-4=(3x-2)(3x+2)$$

$$3x^2-13x-10=(3x+2)(x-5)$$

So the first product becomes

$$\frac{1}{3x-2} \times \frac{(3x-2)(3x+2)}{(3x+2)(x-5)}$$

Cancel common factors:

$$\frac{1}{x-5}$$

Now subtract:

$$\frac{1}{x-5} - \frac{7}{x-1}$$

Use the common denominator $(x-5)(x-1)$:

$$\frac{(x-1)-7(x-5)}{(x-5)(x-1)}$$

$$= \frac{(x-1-7x+35)}{(x-5)(x-1)}$$

$$= \frac{(34-6x)}{(x-5)(x-1)}$$

Answer: $\boxed{\frac{(34-6x)}{(x-5)(x-1)}}$

4MA1-2H Q22 Q22

7 marks

- 22 $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,
find the coordinates of the points B and D .

(..... ,)

(..... ,)

WORKING SPACE

4MA1-2H Q22 Q22

7 marks

- 22 $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,
find the coordinates of the points B and D .

(..... ,)

(..... ,)

Chapter: Coordinate Geometry

7 marks

Q22 – Chapter: Coordinate Geometry

Family: Determine coordinates from geometric information • Variant: Kite or symmetric vertex coordinates

Method

The gradient of AM is $(2 - 4)/(0 - (-3)) = -2/3$.

Why: Use the two given coordinates.

Method

The kite diagonals are perpendicular, so BD has gradient $3/2$ and equation $y = (3/2)x + 2$.

Why: Use the negative reciprocal and point M.

Method

For B or D, $(x + 3)^2 + (y - 4)^2 = 6.5^2$.

Why: Each point is 6.5 units from A.

Method

Substitute $y = (3/2)x + 2$: $(x + 3)^2 + ((3/2)x - 2)^2 = 42.25$.

Why: Combine the line and distance conditions.

Method

Expanding gives $3.25x^2 + 13 = 42.25$.

Why: Expand and collect; the linear x terms cancel.

Method

Thus $x^2 = 9$, so $x = 3$ or $x = -3$; the corresponding y -values are 6.5 and -2.5.

Why: Solve the quadratic and substitute into the line.

Method

Since B has positive x -coordinate, $B = (3, 6.5)$ and $D = (-3, -2.5)$.

Why: Use the final labelling condition.

Final answer

B: $B = (3, 6.5)$

Final answer

D: $D = (-3, -2.5)$

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$

J is the point with coordinates $(j, 15)$ where $j < 0$

K is the point with coordinates $(6, k)$

M is the midpoint of JK .

The gradient of HM is 2

Find the value of j and the value of k .

$j = \dots\dots\dots$

$k = \dots\dots\dots$

Worked solution

January 2020 | Question 22 | 6 marks

Family: Determine coordinates from geometric information • Variant: Coordinates from isosceles and gradient constraints

Method / working

1. $M = ((j+6)/2, (15+k)/2)$.
2. Gradient $HM=2$ gives $(k+13)/(j+14)=2$, so $k=2j+15$.
3. $JK^2=80$ gives $(6-j)^2+(k-15)^2=80$.
4. Substitute k : $5j^2-12j-44=0$.
5. The condition $j<0$ selects $j=-2$.
6. Then $k=2(-2)+15=11$.

Final answer: $j=-2$, $k=11$

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$

B has coordinates $(1, 10)$

C has coordinates $(19, a)$

D has coordinates (b, c)

(a) Work out the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

..... cm

(3)

Worked solution

January 2021 | Question 18 | 4 marks

Family: Determine coordinates from geometric information • Variant: Rectangle or parallelogram missing vertex

Method / working

1. $AB=(12,5)$, so $b=14$ from the rectangle translation.
2. Gradient $AB=12/5$, hence gradient $BC=-5/12$.
3. Use $C=(19,a)$ to obtain $a=2.5$ and then $D=(14,-9.5)$.
4. State all three coordinates values.

Final answer: $a=2.5$, $b=14$, $c=-9.5$

23 $ABCD$ is a kite.

$$AB = AD \text{ and } CB = CD$$

The point B has coordinates $(k, 1)$ where k is a negative constant.

The point D has coordinates $(8, 7)$

The straight line L passes through the points B and D

The straight line L is parallel to the line with equation $5y - 3x = 6$

Find an equation of AC

Give your answer in the form $px + qy = r$ where p , q and r are integers.

Show your working clearly.



(Total for Question 23 is 6 marks)

Answer reference | January 2023 | Question 23

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

23	$y = \frac{3}{5}x \left(+ \frac{6}{5} \right)$ or $y = 0.6x(+1.2)$ or (gradient =) $\frac{3}{5}$ or 0.6 oe		6	M1 for correct gradient which may be seen in an equation. Condone $\frac{3}{5}x$ or $0.6x$
	$k = -2$			B1 for $k = -2$
	$\left(\frac{-2+8}{2}, \frac{1+7}{2} \right)$ oe or (3, 4)			M1 for finding the midpoint (use of their k where $k < 0$)
	$\frac{3}{5}m = -1$ or $(m =) -\frac{5}{3}$			M1ft their gradient for use of $m_1 \times m_2 = -1$ Allow $-\frac{5}{3} = -1.67$ or better
	$"4" = -\frac{5}{3}"x"3" + c$ or $c = 9$ or $y - "4" = -\frac{5}{3}(x - "3")$			M1 dep on M3
	Working required	$5x + 3y = 27$		A1 allow equation in any form where p, q and r are integers
Total 6 marks				

Worked solution

January 2023 | Question 23 | 6 marks

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

Official final mark-scheme pages are embedded immediately after this question.

4MA1-2H Q18

2 marks

18 The straight line **P** has equation $5y + 2x = 7$

Find the gradient of a straight line that is perpendicular to **P**

WORKING SPACE

4MA1-2H Q18

2 marks

18 The straight line **P** has equation $5y + 2x = 7$

Find the gradient of a straight line that is perpendicular to **P**

Perpendicular gradient

2 marks

Gradient

$$5y + 2x = 7 \Rightarrow m = -2/5; \quad m_{\perp} = 5/2.$$

Final answer

$$\frac{5}{2}$$

19 (a) Write down an equation of a line that is parallel to the line with equation $y = 7 - 4x$

(1)

Worked solution

January 2020 | Question 19 | 1 marks

Family: Construct a straight-line equation • Variant: Parallel line through a point

Method / working

1. A parallel line has the same gradient as $y=7-4x$; using the official diagram convention the required parallel gradient is $+4$, so $y=4x$ is valid.

Final answer: $y=4x$

- 20** The straight line **L** passes through point $A(-6, 2)$ and point $B(5, 3)$
The straight line **M** is perpendicular to **L** and passes through the midpoint of A and B .
The line **M** intersects the line $x = -1$ at point C .

Calculate the area of triangle ABC .

Worked solution

November 2021 | Question 20 | 7 marks

Family: Construct a straight-line equation • Variant: Perpendicular line through a point or midpoint

Method / working

1. $A=(-6,2)$, $B=(5,3)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. The midpoint of (AB) is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $\left(\frac{-6+5}{2}, \frac{2+3}{2}\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. =

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. $\left(-\frac{1}{2}, \frac{5}{2}\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. The gradient of (AB) is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $\frac{3-2}{5-(-6)} = \frac{1}{11}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. So the perpendicular gradient is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. -11

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. Line (M) passes through $\left(-\frac{1}{2}, \frac{5}{2}\right)$, so

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. $y - \frac{5}{2} = -11\left(x + \frac{1}{2}\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. At point (C) , $(x=-1)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. $y - \frac{5}{2} = -11\left(-\frac{1}{2}\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. $y - \frac{5}{2} = \frac{11}{2}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

15. $y=8$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

16. So

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

17. $C=(-1,8)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

18. Now use vectors from (A) :

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

19. $\vec{AB}=(11,1)$, $\vec{AC}=(5,6)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

20. Area of triangle (ABC) is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

21. $\frac{1}{2} |11(6)-1(5)|$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

22. $=\frac{1}{2}(61)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

23. $=30.5$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Worked solution (continued)

24. Answer: $\sqrt{30.5}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

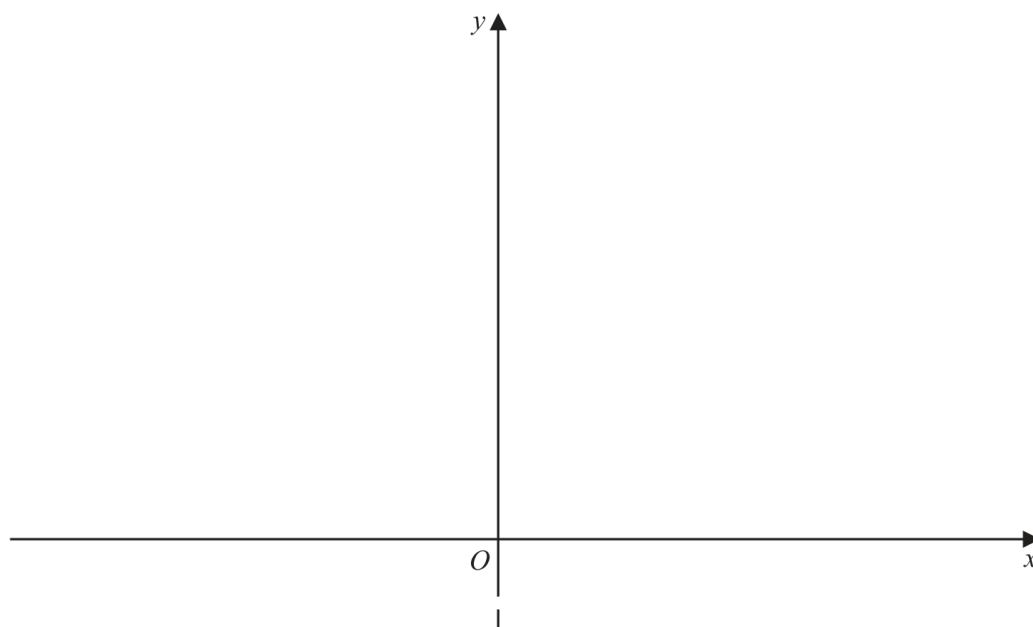
Final answer: 30.5

4MA1-2H Q25

3 marks

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y -axis.



(3)

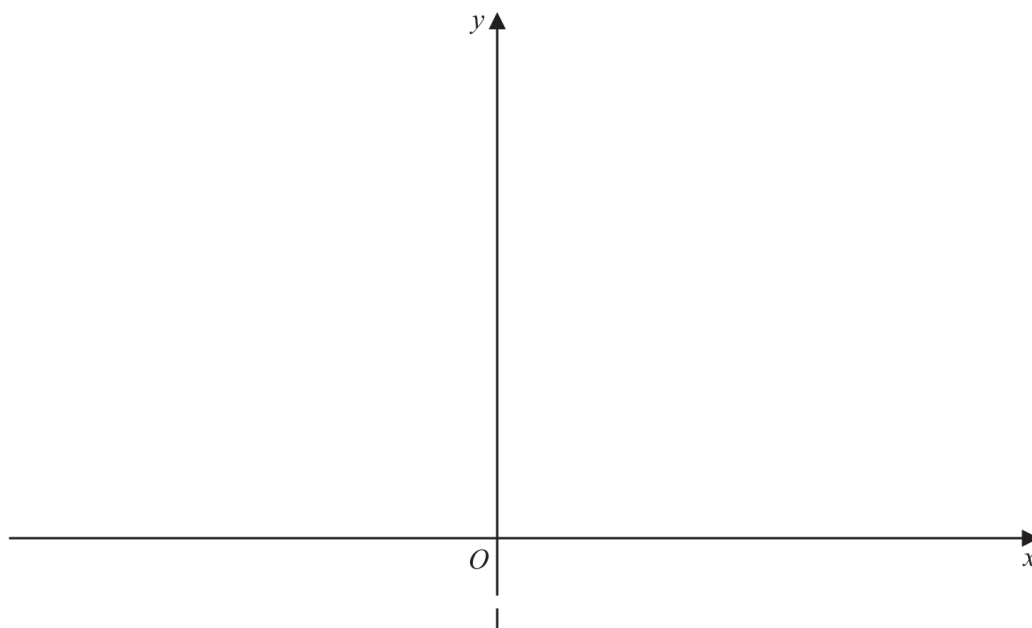
WORKING SPACE

4MA1-2H Q25

3 marks

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y-axis.



(3)

Sketch a quadratic from completed-square form

3 marks

Turning point

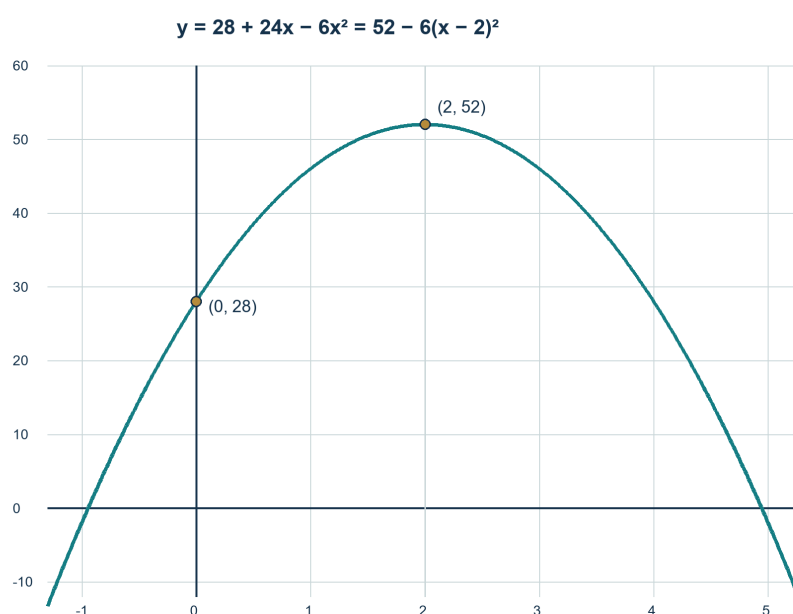
$$y = 52 - 6(x - 2)^2 \Rightarrow \text{maximum turning point } (2, 52).$$

y-intercept

$$x = 0 \Rightarrow y = 28, \text{ so the graph crosses the } y\text{-axis at } (0, 28).$$

Shape

The coefficient of $(x - 2)^2$ is negative, so draw a symmetric downward parabola about $x = 2$.



Solved reference diagram

Final answer

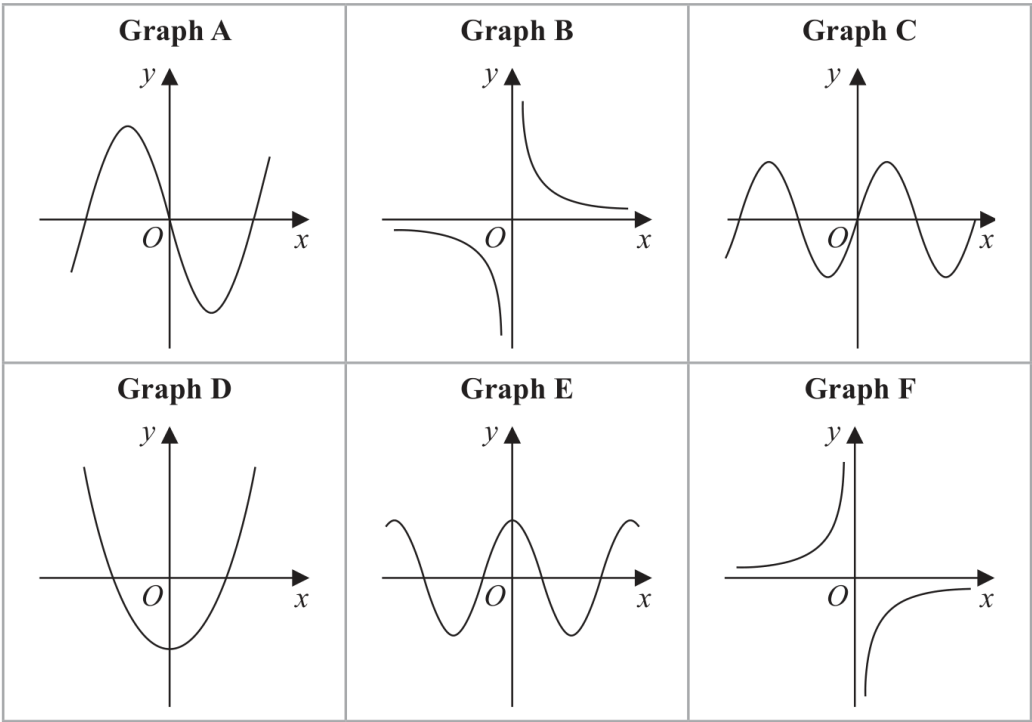
Downward parabola with turning point $(2, 52)$ and y -intercept $(0, 28)$.

Dependency check Sequenced immediately after WU027; the printed equation is repeated, so the crop is self-contained.

4MA1-2H Q18

3 marks

18 Here are 6 graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \sin x$	
$y = -\frac{3}{x}$	
$y = 4x^3 - 5x$	

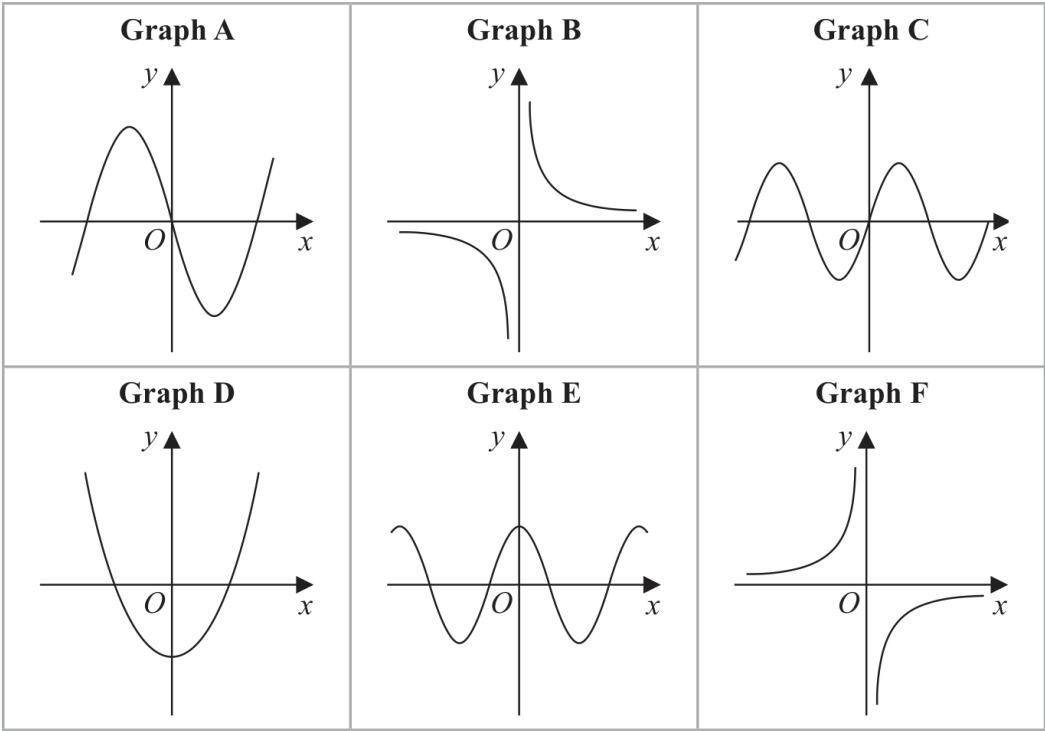
Continue your method**3 marks**

WORKING SPACE

4MA1-2H Q18

3 marks

18 Here are 6 graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \sin x$	
$y = -\frac{3}{x}$	
$y = 4x^3 - 5x$	

Recognise function graph families

3 marks

Match the sine graph

$y = \sin x$ passes through the origin and oscillates : Graph C.

Match the reciprocal graph

$y = -3/x$ has positive branch for $x < 0$ and negative branch for $x > 0$: Graph F.

Match the cubic

$y = 4x^3 - 5x$ is an odd cubic with negative central gradient : Graph A.

Final answer

$$\sin x \rightarrow C, \quad -3/x \rightarrow F, \quad 4x^3 - 5x \rightarrow A$$

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$

B has coordinates $(1, 10)$

C has coordinates $(19, a)$

D has coordinates (b, c)

(a) Work out the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

..... cm

(3)

Worked solution

January 2021 | Question 18 | 3 marks

Family: Find perimeter of polygonal shapes • Variant: Find perimeter of a simple triangle or rectangle

Method / working

1. $AB = \sqrt{5^2 + 12^2} = 13 \text{ cm.}$
2. $BC = \sqrt{18^2 + 7.5^2} = 19.5 \text{ cm.}$
3. $\text{Perimeter} = 2(13 + 19.5) = 65 \text{ cm.}$

Final answer: 65 cm

20

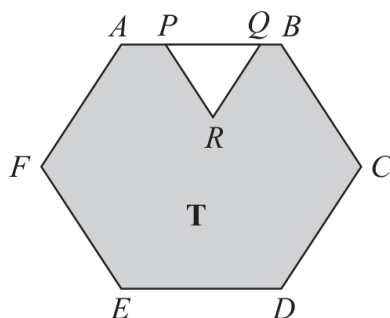


Diagram **NOT**
accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3} \text{ cm}^2$

work out the length of PQ .

..... cm

Worked solution

January 2021 | Question 20 | 4 marks

Family: Use rectangle and triangle area • Variant: Find composite or shaded polygonal area

Method / working

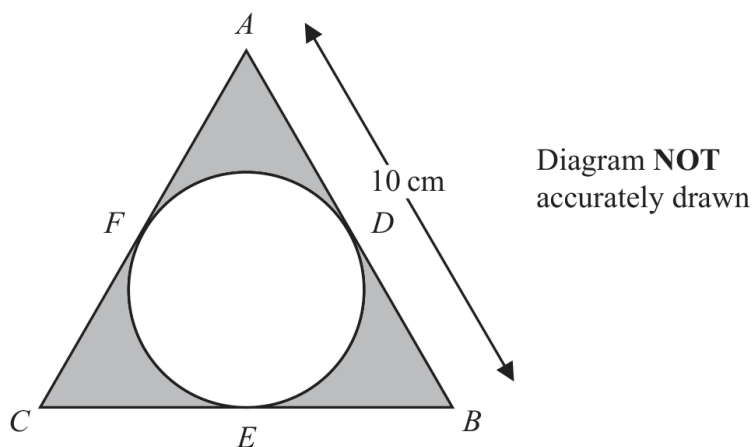
1. Let $PQ=x$, so $AB=1.5x$.
2. Write the equilateral triangle and regular-hexagon areas in terms of x .
3. Set shaded area equal to $72\sqrt{3}$ and solve the resulting quadratic in x^2 .
4. The positive length is $PQ=4.8$ cm.

Final answer: 4.8 cm

4MA1-2H Q20 whole

4 marks

- 20 The diagram shows equilateral triangle ABC with sides of length 10 cm.
A circle is drawn inside the triangle.



D , E and F are points on the circle.

ADB , BEC and CFA are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.

Give your answer correct to 3 significant figures.

..... cm^2

WORKING SPACE

Continue your method

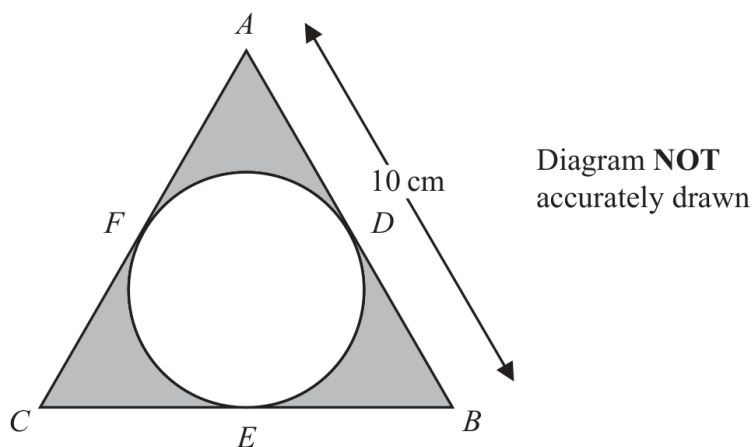
4 marks

WORKING SPACE

4MA1-2H Q20 whole

4 marks

- 20 The diagram shows equilateral triangle ABC with sides of length 10 cm.
A circle is drawn inside the triangle.



D , E and F are points on the circle.

ADB , BEC and CFA are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

..... cm^2

Shaded area between an equilateral triangle and its incircle

4 marks

whole – Shaded area between an equilateral triangle and its incircle**Find the triangle area**

$$A_{\triangle} = \frac{1}{2}(10)(10) \sin 60^{\circ} = 25\sqrt{3}.$$

Find the inradius

$$r = \frac{10\sqrt{3}}{6} = \frac{5\sqrt{3}}{3}.$$

Subtract the circle area

$$A_{\text{shaded}} = 25\sqrt{3} - \pi r^2 = 25\sqrt{3} - \frac{25\pi}{3}.$$

Round

$$A_{\text{shaded}} = 17.121 \dots \approx 17.1 \text{ cm}^2.$$

Final answer17.1 cm²

4MA1-1H Q22

6 marks

22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

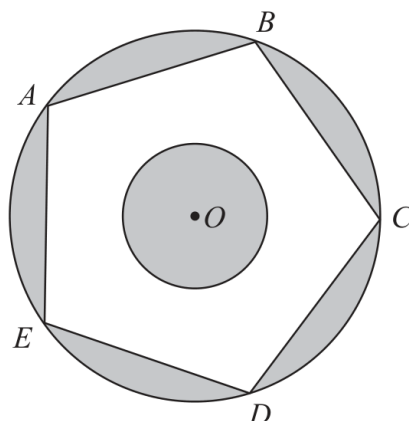


Diagram **NOT**
accurately drawn

A , B , C , D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

$$\text{shaded area} = \text{unshaded area}$$

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

..... cm

Continue your method**6 marks**

WORKING SPACE

4MA1-1H Q22

6 marks

- 22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

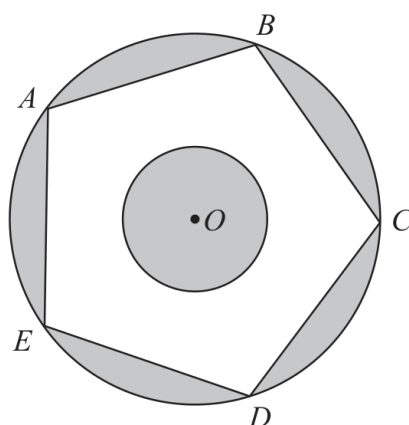


Diagram **NOT**
accurately drawn

A , B , C , D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

$$\text{shaded area} = \text{unshaded area}$$

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

..... cm

Equal shaded and unshaded areas

6 marks

Find the circumradius

$$R = \frac{4}{\sin 36^\circ} = 6.805 \dots$$

Find the two main areas

$$A_{\text{circle}} = \pi R^2 = 145.489 \dots, \quad A_{\text{pentagon}} = 5 \left(\frac{1}{2} R^2 \sin 72^\circ \right) = 110.111 \dots$$

Use the equal-area condition

$$A_{\text{circle}} - A_{\text{pentagon}} + \pi r^2 = A_{\text{pentagon}} - \pi r^2.$$

Solve

$$r = \sqrt{\frac{2A_{\text{pentagon}} - A_{\text{circle}}}{2\pi}} = 3.44 \dots$$

Final answer

3.45 cm

4MA1-1H Q16 M01

4 marks

16 $OAPB$ is a sector of a circle, centre O

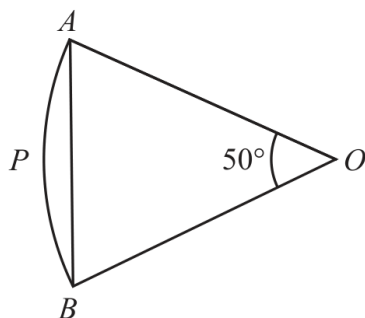


Diagram **NOT**
accurately drawn

Angle $AOB = 50^\circ$

Area of triangle $OAB = 120 \text{ cm}^2$

Work out the area of the sector $OAPB$

Give your answer correct to 3 significant figures.

..... cm^2

WORKING SPACE

Find the area of a sector

4 marks

M01 – Find the area of a sector

Use the triangle area

$$120 = \frac{1}{2}r^2 \sin 50^\circ \Rightarrow r^2 = 240 / \sin 50^\circ.$$

Find r

$$r = 17.7002 \dots$$

Use the sector fraction

$$A = \frac{50}{360} \pi r^2 = 136.7 \dots$$

Round

$$A = 137 \text{ cm}^2 \text{ (3 s.f.)}$$

Final answer

$$137 \text{ cm}^2$$

4MA1-2H Q25 M01

5 marks

25 The diagram shows an equilateral triangle ABC and a circle with centre O

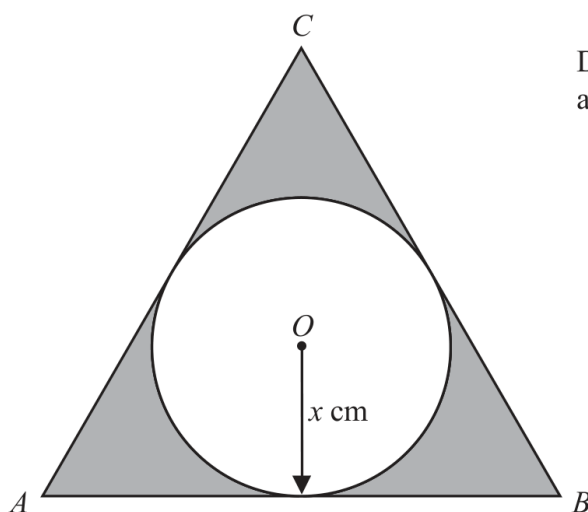


Diagram **NOT**
accurately drawn

AB , BC and CA are tangents to the circle.

The radius of the circle is x cm

The total area, in cm^2 , of the regions shown shaded in the diagram is nx^2

Find the value of n

Give your answer correct to 3 significant figures.

4MA1-2H Q25 M01

5 marks

 $n = \dots\dots\dots$

WORKING SPACE

Area of shaded regions in an equilateral triangle

5 marks

M01 – Area of shaded regions in an equilateral triangle

Find the side length

The inradius is x , so the side is $2\sqrt{3}x$.

Find triangle area

$$A_{\text{triangle}} = \frac{1}{2}(2\sqrt{3}x)(3x) = 3\sqrt{3}x^2.$$

Subtract the circle

$$A_{\text{shaded}} = (3\sqrt{3} - \pi)x^2.$$

Read n

$$n = 3\sqrt{3} - \pi = 2.0548\dots$$

Round

$$n = 2.05 \text{ (3 s.f.)}.$$

Final answer

2.05

4MA1-1H Q16

4 marks

16 The diagram shows a circle with centre O

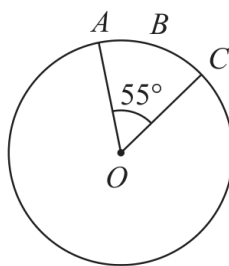


Diagram **NOT**
accurately drawn

A , B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

..... cm^2

Continue your method

4 marks

WORKING SPACE

4MA1-1H Q16

4 marks

16 The diagram shows a circle with centre O

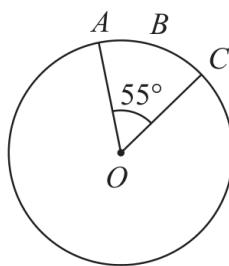


Diagram **NOT**
accurately drawn

A , B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

..... cm^2

Chapter: Circles, Arcs & Sectors

4 marks

Q16 – Chapter: Circles, Arcs & Sectors

Family: Use arc and sector measures • Variant: Recover radius or central angle from an arc/sector measure

Method

$$5 = \left(\frac{55}{360}\right) \times 2\pi r, \text{ so } r = 5 \times 360 / (55 \times 2\pi).$$

Why: Use the arc-length formula $s = (\theta/360)2\pi r$.

Method

$$r \approx 5.207 \text{ cm.}$$

Why: Evaluate the radius.

Method

$$\text{Area} = \pi r^2 \approx 85.2 \text{ cm}^2.$$

Why: Use πr^2 .

Method

To 1 decimal place, the area is 85.2 cm^2 .

Why: Round as requested.

Final answer

area: 85.2 cm^2

- 21 The diagram shows the cross section of a circular water pipe.

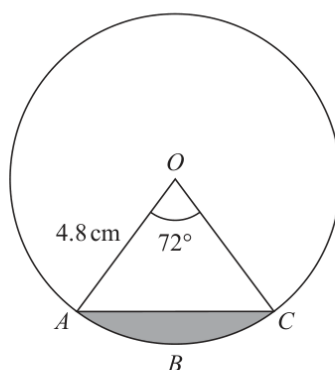


Diagram **NOT**
accurately drawn

OAC is a sector of the circle, centre O

The shaded region in the diagram represents the water flowing in the pipe.

The water flows at 14 cm/s in the pipe.

Work out the volume of water that has flowed through the pipe in 3 minutes.
Give your answer in cm^3 correct to 3 significant figures.



..... cm³

(Total for Question 21 is 5 marks)

Answer reference | January 2023 | Question 21

Family: Resolve compound and shaded circle regions • Variant: Find a circular segment area or perimeter

21	$\pi \times 4.8^2 \times \frac{72}{360}$ (= 14.4(76...)) oe		5	M1 for finding the area of the sector
	$\frac{1}{2} \times 4.8^2 \times \sin 72$ (= 10.9(56...) or 11) oe or $\frac{1}{2} \times 5.6(4...) \times 3.8(8...) \text{ oe}$			M1 for finding the area of the triangle (Allow use of cosine rule/sine rule/SOHCAHTOA/Pythagoras to find AC (5.6(427.8...)) and OM (3.8(8328...)) where M is the midpoint of AC)
	"14.4(76...)" – "10.9(56...)" (= 3.520...)			M1 for finding the shaded area with all figures from correct working
	"3.5(20...)" $\times 14 \times 3 \times 60$ "3.5(20...)" $\times 2520$			M1
	<i>Award marks within the range from correct working</i>	8870		A1 accept 8820 – 8950 from correct working
				Total 5 marks

Worked solution

January 2023 | Question 21 | 5 marks

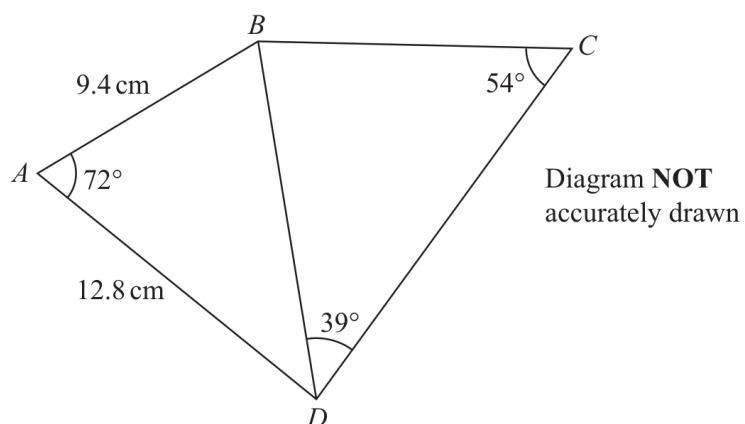
Family: Resolve compound and shaded circle regions • Variant: Find a circular segment area or perimeter

Official final mark-scheme pages are embedded immediately after this question.

4MA1-1H Q20

5 marks

20



Work out the length of BC
Give your answer correct to 3 significant figures.

..... cm

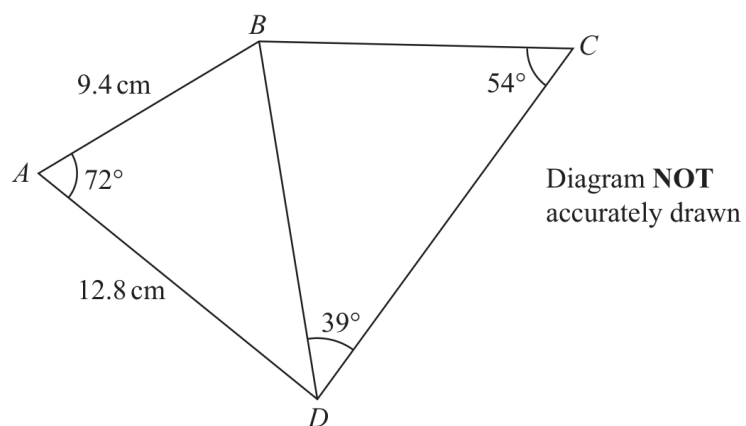
(Total for Question 20 is 5 marks)

WORKING SPACE

4MA1-1H Q20

5 marks

20



Work out the length of BC
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

Cosine rule followed by sine rule

5 marks

Find the shared side

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8) \cos 72^\circ, \quad BD = 13.335597 \dots$$

Use the sine rule

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ} \Rightarrow BC = \frac{13.335597 \dots \sin 39^\circ}{\sin 54^\circ} = 10.3735 \dots$$

Final answer

$$BC = 10.4 \text{ cm (3 s.f.)}$$

Dependency check Retain the unrounded value of BD for the second triangle.

4MA1-2H Q22

5 marks

22 The diagram shows a triangle ABC and a flagpole BF

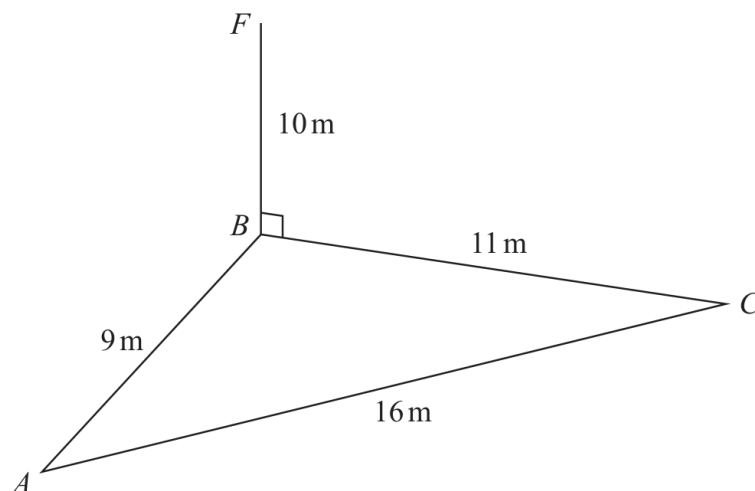


Diagram **NOT**
accurately drawn

A , B and C are points on horizontal ground.

BF is vertical.

$$AB = 9 \text{ m} \quad BC = 11 \text{ m} \quad AC = 16 \text{ m} \quad BF = 10 \text{ m}$$

D is the point on AC such that angle $BDC = 90^\circ$

Work out the size of the angle of elevation of the point F from the point D
Give your answer correct to one decimal place.

○

Continue your method

5 marks

WORKING SPACE

4MA1-2H Q22

5 marks

22 The diagram shows a triangle ABC and a flagpole BF

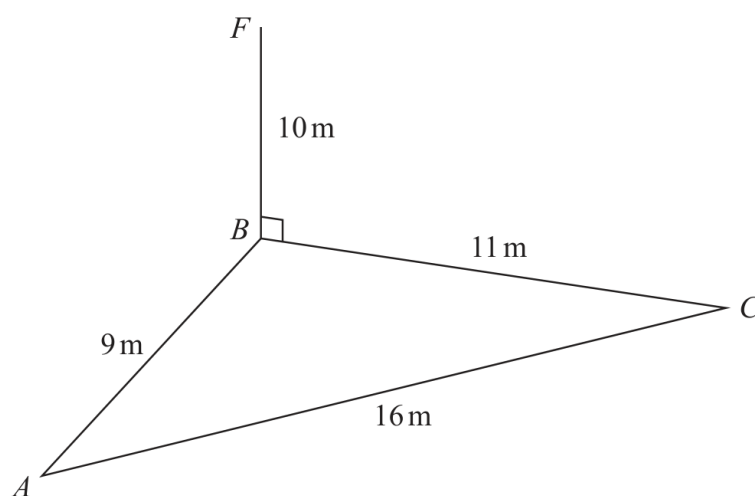


Diagram **NOT**
accurately drawn

A , B and C are points on horizontal ground.

BF is vertical.

$$AB = 9\text{ m} \quad BC = 11\text{ m} \quad AC = 16\text{ m} \quad BF = 10\text{ m}$$

D is the point on AC such that angle $BDC = 90^\circ$

Work out the size of the angle of elevation of the point F from the point D

Give your answer correct to one decimal place.

○

Angle of elevation from a perpendicular foot

5 marks

Find the ground angle

$$\cos \angle BCA = \frac{11^2 + 16^2 - 9^2}{2(11)(16)} = \frac{37}{44} \Rightarrow \angle BCA \approx 32.763^\circ.$$

Find BD

$$BD = 11 \sin 32.763^\circ \approx 5.95 \text{ m.}$$

Use the vertical flagpole

$$\tan \angle FDB = \frac{BF}{BD} = \frac{10}{5.95}.$$

Evaluate

$$\angle FDB \approx 59.2^\circ.$$

Final answer

$$59.2^\circ$$

4MA1-1H Q18 Q18

5 marks

18 Here is triangle ABC

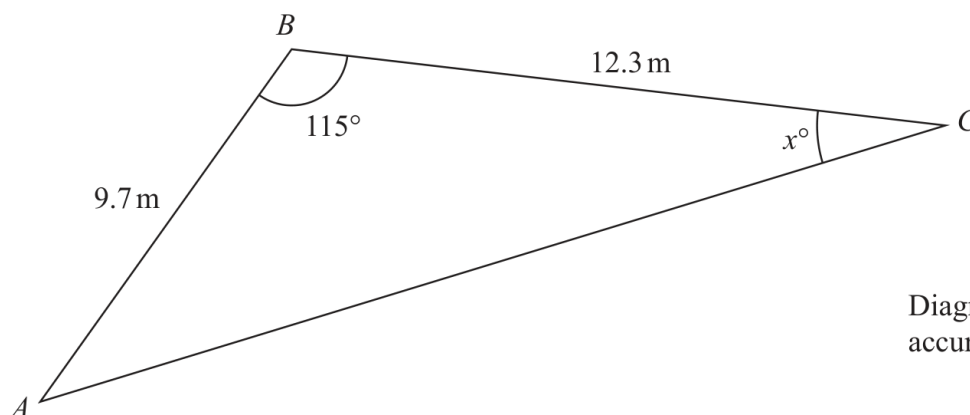


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

$x =$

Continue your method**5 marks**

WORKING SPACE

4MA1-1H Q18 Q18

5 marks

18 Here is triangle ABC

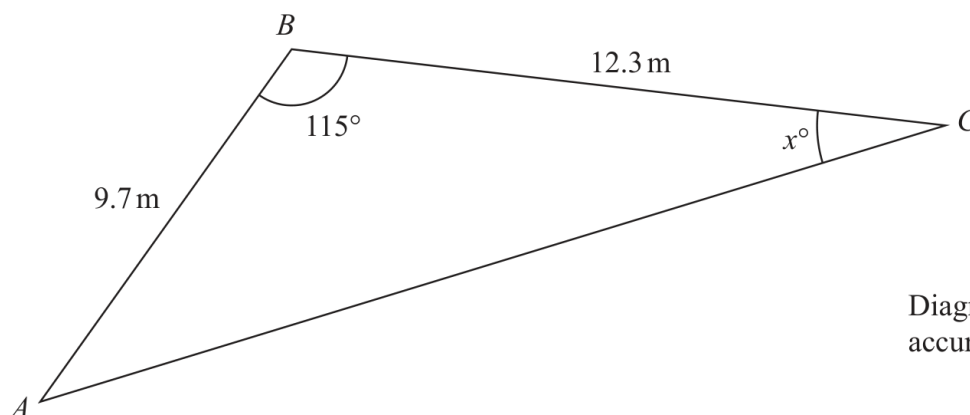


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

$x =$

Chapter: Sine, Cosine Rule & Area of Triangles

5 marks

Q18 – Chapter: Sine, Cosine Rule & Area of Triangles

Family: Use the sine rule • Variant: Find an acute angle using the sine rule

MethodBy the cosine rule, $AC^2 = 9.7^2 + 12.3^2 - 2(9.7)(12.3)\cos(115 \text{ degrees})$.*Why: The two known sides enclose the known 115-degree angle.***Method** $AC^2 = 346.2... , \text{ so } AC = 18.606... \text{ m.}$ *Why: Evaluate the cosine-rule expression without premature rounding.***Method**Using the sine rule, $\sin(x)/9.7 = \sin(115 \text{ degrees})/18.606... .$ *Why: Side AB = 9.7 is opposite angle x, and AC is opposite 115 degrees.***Method**Therefore $\sin(x) = 9.7 \sin(115 \text{ degrees})/18.606... = 0.472... .$ *Why: Rearrange the sine-rule equation for $\sin(x)$.***Method** $x = \sin^{-1}(0.472...) = 28.2 \text{ degrees to 3 significant figures.}$ *Why: Take the inverse sine in degree mode and round once at the end.***Final answer****angle:** $x = 28.2 \text{ degrees}$

4MA1-2H Q23

6 marks

23 The diagram shows triangle PQR

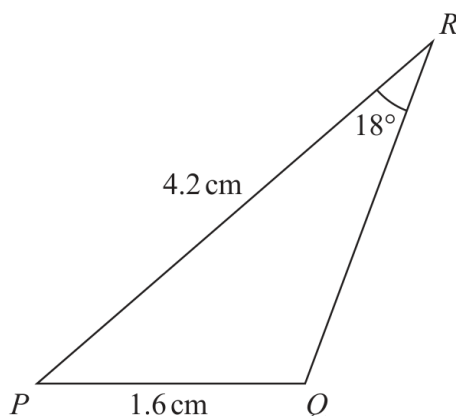


Diagram **NOT**
accurately drawn

$$PQ = 1.6 \text{ cm}$$

$$PR = 4.2 \text{ cm}$$

$$\text{Angle } PRQ = 18^\circ$$

Given that angle PQR is obtuse,

work out the area of triangle PQR

Give your answer correct to 3 significant figures.

..... cm^2

Continue your method**6 marks**

WORKING SPACE

4MA1-2H Q23

6 marks

23 The diagram shows triangle PQR

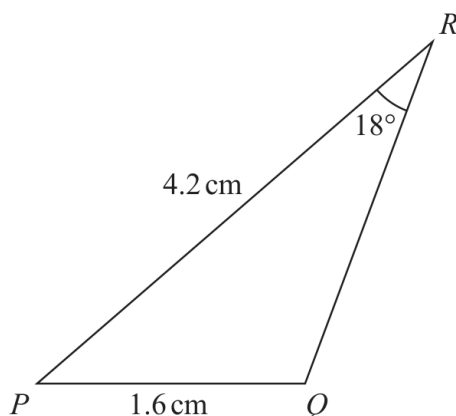


Diagram **NOT**
accurately drawn

$$PQ = 1.6 \text{ cm}$$

$$PR = 4.2 \text{ cm}$$

$$\text{Angle } PRQ = 18^\circ$$

Given that angle PQR is obtuse,

work out the area of triangle PQR

Give your answer correct to 3 significant figures.

..... cm^2

Chapter: Sine, Cosine Rule & Area of Triangles

6 marks

Q23 – Chapter: Sine, Cosine Rule & Area of Triangles

Family: Use the sine rule • Variant: Resolve the ambiguous sine-rule angle

MethodUsing the sine rule, $\sin Q/4.2 = \sin 18^\circ/1.6$, giving $Q = 54.2^\circ$ or 125.8° .*Why: The sine equation has an acute and an obtuse possibility.***Method**The question says Q is obtuse, so $Q = 125.8^\circ$.*Why: Select the permitted branch.***Method** $P = 180 - 125.8 - 18 = 36.2^\circ$.*Why: Angles in a triangle total 180° .***Method**

The corresponding side RQ is 3.0585... cm.

*Why: Use the cosine rule or the sine rule after finding P.***Method** $\text{Area} = \frac{1}{2} \times 4.2 \times 1.6 \times \sin 36.2^\circ$.*Why: Use two sides and the included angle.***Method** $\text{Area} \approx 1.98 \text{ cm}^2$ (3 s.f.).*Why: Round the calculator value.***Final answer**area: 1.98 cm^2

- 16 Here is a shape formed from two triangles ABC and CDE
 ACD and BCE are straight lines.

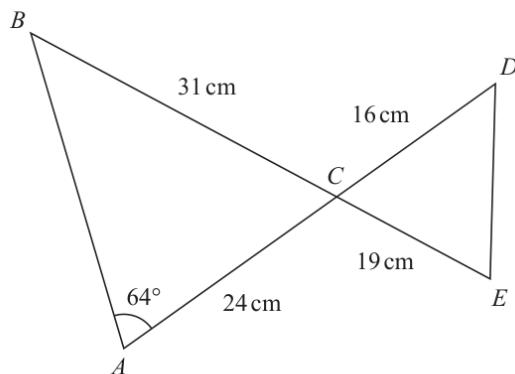


Diagram **NOT**
 accurately drawn

$$AC = 24 \text{ cm} \quad BC = 31 \text{ cm} \quad CE = 19 \text{ cm} \quad CD = 16 \text{ cm}$$

$$\text{Angle } BAC = 64^\circ$$

Work out the length of DE

Give your answer correct to 3 significant figures.



January 2023 | 1H | Question 16 | 5 marks

Chapter OL-T49: Sine, Cosine Rule & Area of Triangles

..... cm

(Total for Question 16 is 5 marks)

Answer reference | January 2023 | Question 16

Family: Use the sine rule • Variant: Find a side using the sine rule

16	$\frac{\sin ABC}{24} = \frac{\sin 64}{31}$ oe		5	M1
	$(ABC =) \sin^{-1}\left(\frac{24 \times \sin 64}{31}\right) (= 44. \dots)$			M1
	$180 - "44. \dots" - 64 (= 71.9 \dots)$			M1 accept 72
	$(DE^2 =) 16^2 + 19^2 - 2 \times 16 \times 19 \times \cos "71.9 \dots"$ or $(DE =) \sqrt{16^2 + 19^2 - 2 \times 16 \times 19 \times \cos "71.9 \dots"}$ or $(DE =) \sqrt{617 - 181.8 \dots}$ or $\sqrt{428.166 \dots}$			M1 for DE^2 or DE
	<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	20.7		A1 awrt 20.7
				Total 5 marks

Worked solution

January 2023 | Question 16 | 5 marks

Family: Use the sine rule • Variant: Find a side using the sine rule

Official final mark-scheme pages are embedded immediately after this question.

4MA1-1H Q21

5 marks

- 21 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$

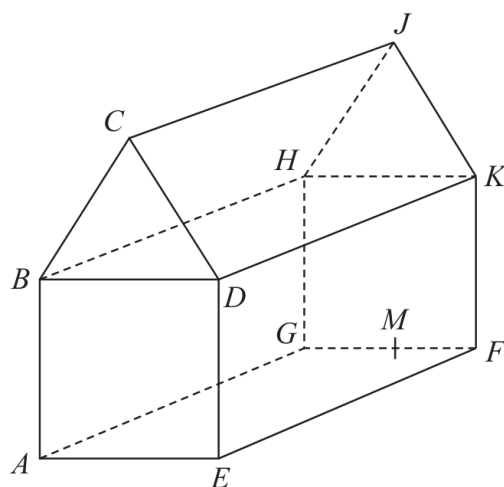


Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where
 $ABDE$ is a square
 BCD is an equilateral triangle

$$EF = 2 \times AE$$

M is the midpoint of GF so that JM is vertical.

$$\text{Angle } MAJ = y^\circ$$

$$\text{Given that } \tan y^\circ = T$$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$T = \dots\dots\dots$

Continue your method**5 marks**

WORKING SPACE

4MA1-1H Q21

5 marks

- 21 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$

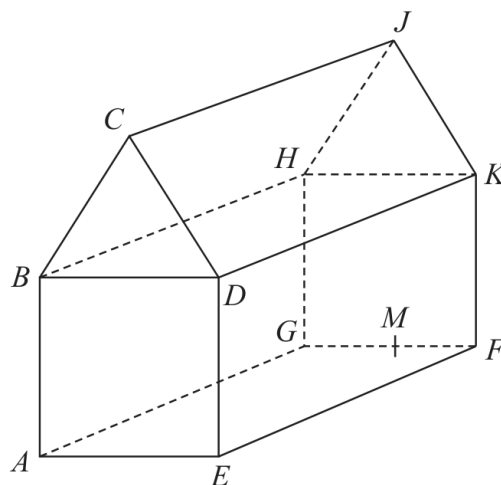


Diagram **NOT**
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$ABCDE$ is a cross section of the prism where
 $ABDE$ is a square
 BCD is an equilateral triangle

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M is the midpoint of GF so that JM is vertical.

$$\text{Angle } MAJ = y^\circ$$

$$\text{Given that } \tan y^\circ = T$$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$T = \dots\dots\dots$

Chapter: 3D Pythagoras & Trigonometry

5 marks

Q21 – Chapter: 3D Pythagoras & Trigonometry

Family: Resolve multi-stage solid geometry • Variant: Use midpoint or symmetry structure in a 3D model

Method

$$AM = \sqrt{(x^2 + (4x)^2)} = x\sqrt{17}.$$

Why: Use Pythagoras in the horizontal base triangle.

Method

$$\text{The equilateral-triangle height is } \sqrt{((2x)^2 - x^2)} = x\sqrt{3}.$$

Why: Use the altitude of the equilateral cross-section.

Method

$$\tan y = (x\sqrt{3} + 2x) / (x\sqrt{17}) = (\sqrt{3} + 2) / \sqrt{17}.$$

Why: Use opposite over adjacent for angle MAJ.

Method

$$\text{Rationalising gives } (\sqrt{3} + 2)\sqrt{17}/17 = (\sqrt{51} + 2\sqrt{17})/17.$$

Why: Multiply numerator and denominator by $\sqrt{17}$.

Method

$$\text{Since } 2\sqrt{17} = \sqrt{68}, T = (\sqrt{51} + \sqrt{68})/17.$$

Why: Write the answer in the requested surd form.

Final answer

$$T: (\sqrt{51} + \sqrt{68})/17$$

22 The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

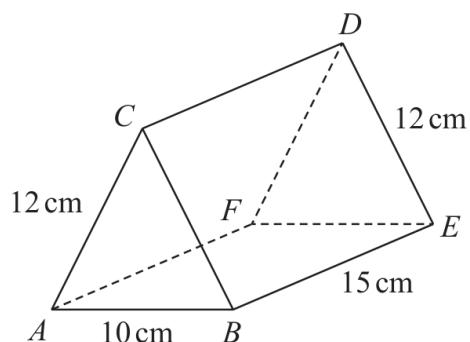


Diagram **NOT**
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm} \quad AB = 10 \text{ cm} \quad BE = 15 \text{ cm}$$

Calculate the size of the angle between AD and the base $ABEF$.
Give your answer correct to 3 significant figures.

..... °

Worked solution

June 2021 | Question 22 | 4 marks

Family: Resolve multi-stage solid geometry • Variant: Use a non-rectangular base before a line-plane angle

Method / working

1. Use the prism diagram to form the right triangles with lengths 15.8, 19.2 and 10.9 cm.
2. Use $\tan(\text{angle}) = 10.9/15.8$ (equivalently a sine/cosine relation).
3. Evaluate the inverse trigonometric function.
4. Round to 1 decimal place: 34.6° .

Final answer: 34.6°

18 The diagram shows cuboid $ABCDEFGH$.

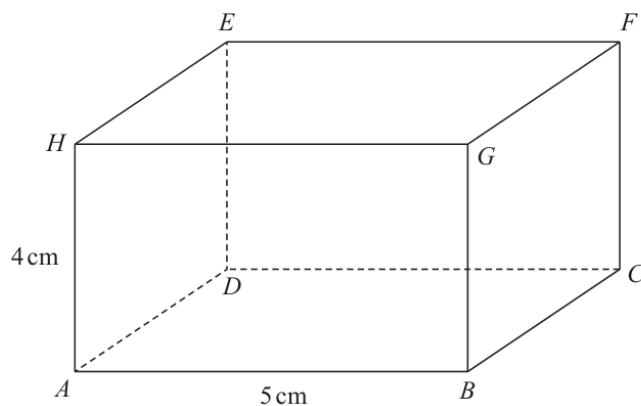


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm}$$

$$AH = 4 \text{ cm}$$

The size of the angle between CH and the plane $ABCD$ is 35°

Calculate the volume of the cuboid.

Give your answer correct to 3 significant figures.

.....cm³

(Total for Question 18 is 5 marks)

Answer reference | May-Nov 2020 | Question 18

Family: Use trigonometry in three dimensions • Variant: Find angle between a line and a plane

18	$\text{eg } \frac{4}{AC} = \tan 35 \text{ oe } \frac{AC}{4} = \tan 55 \text{ oe } \frac{AC}{\sin 55} = \frac{4}{\sin 35} \text{ oe or}$ $CH = \frac{4}{\sin 35} \text{ oe } (= 6.97...) \text{ and } \frac{AC}{6.97} = \cos 35 \text{ oe or}$ $CH = \frac{4}{\sin 35} \text{ oe } (= 6.97...) \text{ and } AC^2 = 6.97^2 - 4^2 \text{ oe}$			M1	A correct trig statement involving AC or trig and then Pythagoras involving AC
	$(AC =) \frac{4}{\tan 35} \text{ oe eg } (AC =) 4 \tan 55 (= 5.71...) \text{ or}$ $(AC =) \frac{4 \sin 55}{\sin 35} \text{ or } "6.97" \times \cos 35 \text{ oe or } (AC =) \sqrt{6.97^2 - 4^2}$			M1	complete method to find AC
	$(BC =) \sqrt{5.71^2 - 5^2} (= 2.76...)$			M1	complete method to find BC
	$4 \times 5 \times "2.76..."$			M1	method to find volume
		55.3	5	A1	accept 55.1 – 55.5
					Total 5 marks

Worked solution

May-Nov 2020 | Question 18 | 5 marks

Family: Use trigonometry in three dimensions • Variant: Find angle between a line and a plane

Official final mark-scheme pages are embedded immediately after this question.

23 The diagram shows a solid prism $ABCDEFGHJ$

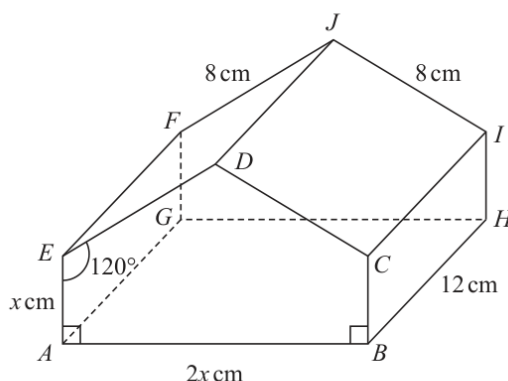


Diagram **NOT** accurately drawn

The prism is such that each cross section is a pentagon where

$$AE = BC = x \text{ cm} \quad AB = 2x \text{ cm} \quad ED = CD = 8 \text{ cm}$$

$$\text{angle } EAB = \text{angle } CBA = 90^\circ \quad \text{angle } AED = \text{angle } BCD = 120^\circ$$

Given that $AG = BH = EF = DJ = CI = 12 \text{ cm}$

calculate the angle that AJ makes with the base $ABHG$ of the prism.

Give your answer correct to 3 significant figures.



(Total for Question 23 is 5 marks)

Answer reference | January 2023 | Question 23

Family: Resolve multi-stage solid geometry • Variant: Use 3D trigonometry to obtain a terminal volume

Q	working	Answer	Mark	Notes
23	$[DN =]8\sin 30 \text{ or } 8\cos 60 (= 4) \text{ oe [where } N \text{ is the midpoint of } EC]$ or $[x =]8\cos 30 \text{ or } 8\sin 60 (= 4\sqrt{3} = 6.928\dots)$ or $2x = \sqrt{8^2 + 8^2 - 2 \times 8 \times 8 \times \cos 120} (= \sqrt{192} = 8\sqrt{3} = 13.85\dots)$		5	M1
	$[DN =]8\sin 30 \text{ or } 8\cos 60 (= 4) \text{ oe eg } \sqrt{8^2 - (4\sqrt{3})^2} (= 4)$ And 1 of $[x =]8\cos 30 \text{ or } 8\sin 60 (= 4\sqrt{3} = 6.928\dots) \text{ oe or}$ $\sqrt{8^2 - "4"^{n2}} (= 4\sqrt{3} = 6.928\dots) \text{ or}$ $2x = \sqrt{8^2 + 8^2 - 2 \times 8 \times 8 \times \cos 120} (= 8\sqrt{3} = 13.85\dots)$			M1 [(JM =) 4 + 4\sqrt{3} implies M2]
	$[AM =]\sqrt{12^2 + ("4\sqrt{3}")^2} (= \sqrt{192} = 8\sqrt{3} = 13.856\dots) \text{ oe}$ [where M is the midpoint of GH]			M1 Clear intention to be AM (not EC)
	$\tan MAJ = \left(\frac{"4" + "4\sqrt{3}"}{8\sqrt{3}} \right) \text{ oe eg } \tan MAJ = \left(\frac{10.928\dots}{13.856\dots} \right)$ if student uses sin or cos, then $AJ = 17.647\dots$ to award marks this must come from a correct method or be correct			M1
	Correct answer scores full marks (unless from obvious incorrect working)	38.3		A1 Accept 38.1 – 38.4 If no marks scored then award SCB2 for $\tan MAJ = \frac{4+x}{\sqrt{x^2 + 12^2}}$ or SCB1 for $AM = \sqrt{x^2 + 12^2}$
				Total 5 marks

Worked solution

January 2023 | Question 23 | 5 marks

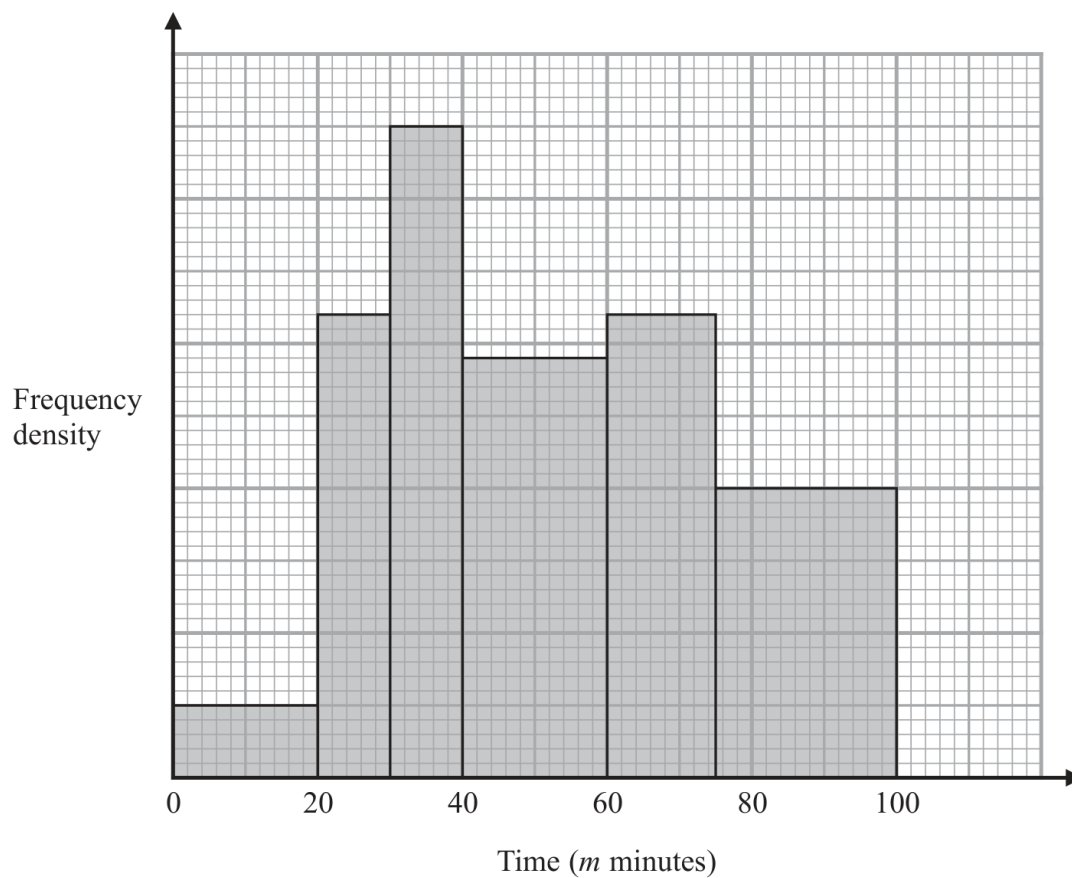
Family: Resolve multi-stage solid geometry • Variant: Use 3D trigonometry to obtain a terminal volume

Official final mark-scheme pages are embedded immediately after this question.

4MA1-1H Q21 Q21

3 marks

- 21 The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$

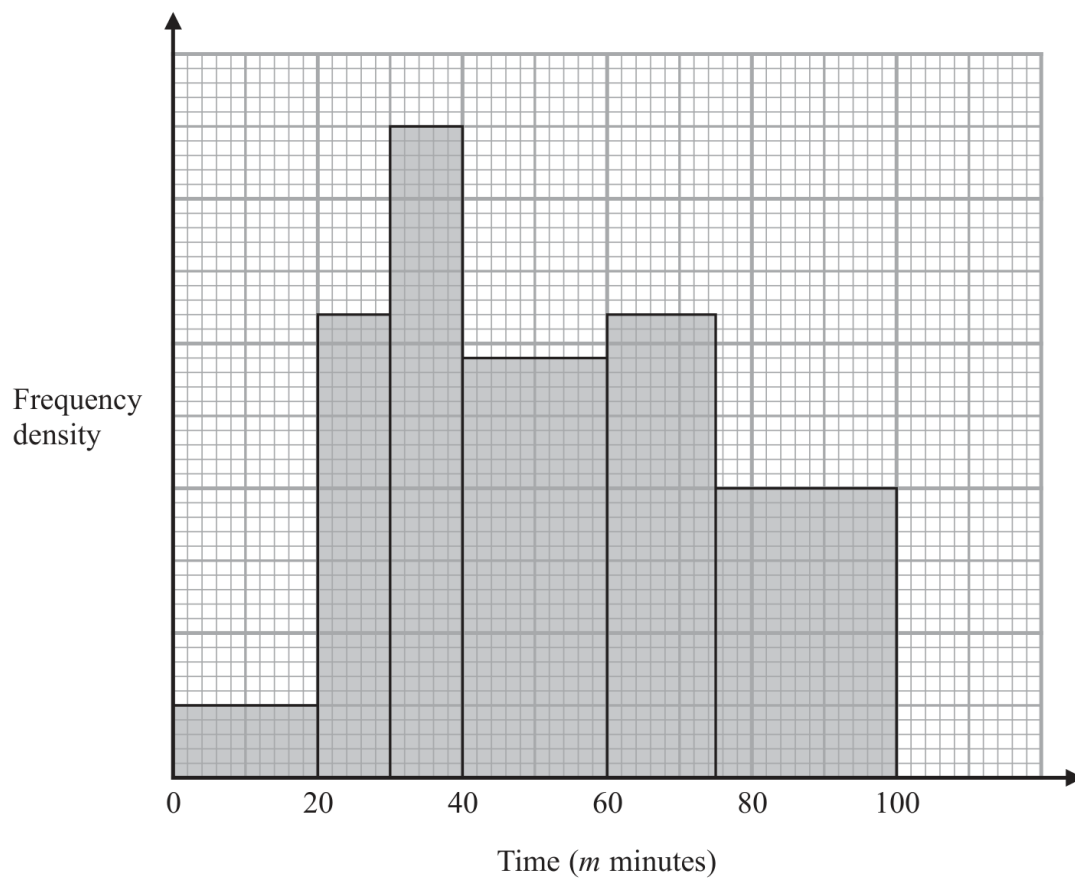
There are 10 children for whom $m \leq 20$

Work out an estimate for the number of children for whom $50 < m \leq 80$

4MA1-1H Q21 Q21

3 marks

- 21 The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$

There are 10 children for whom $m \leq 20$

Work out an estimate for the number of children for whom $50 < m \leq 80$

Chapter: Histograms

3 marks

Q21 – Chapter: Histograms

Family: Interpret histogram areas • Variant: Estimate a partial-class frequency

Method

For $0 < m \leq 20$, frequency density = $10/20 = 0.5$; use this bar to calibrate the vertical scale.

Why: Histogram area equals frequency, and density equals frequency divided by class width.

Method

From the calibrated grid, the densities over 50-60, 60-75, and 75-80 are approximately 2.9, 3.2, and 2.0.

Why: Split the requested interval wherever a histogram class boundary occurs.

Method

Estimated frequency = $10(2.9) + 15(3.2) + 5(2.0) = 29 + 48 + 10 = 87$ children.

Why: Add the areas of the three relevant histogram pieces.

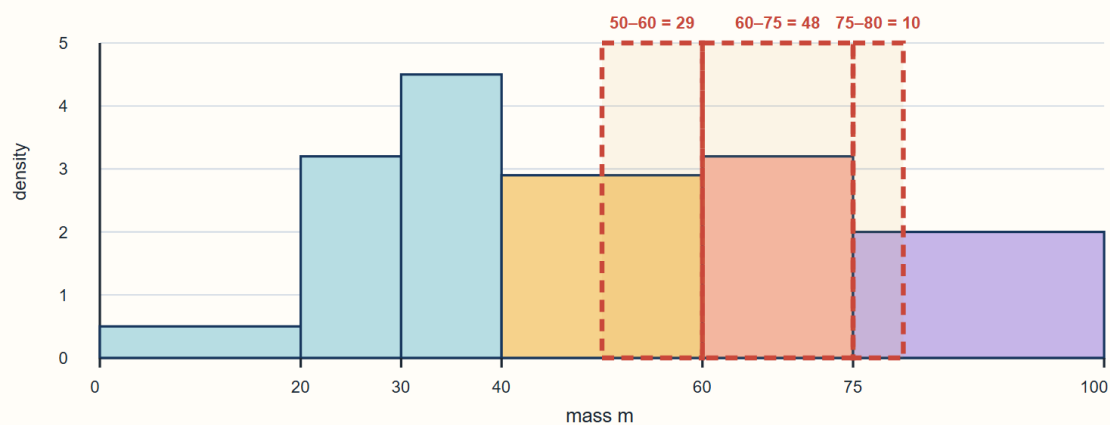
Final answer

estimated children: 87

Chapter: Histograms

3 marks

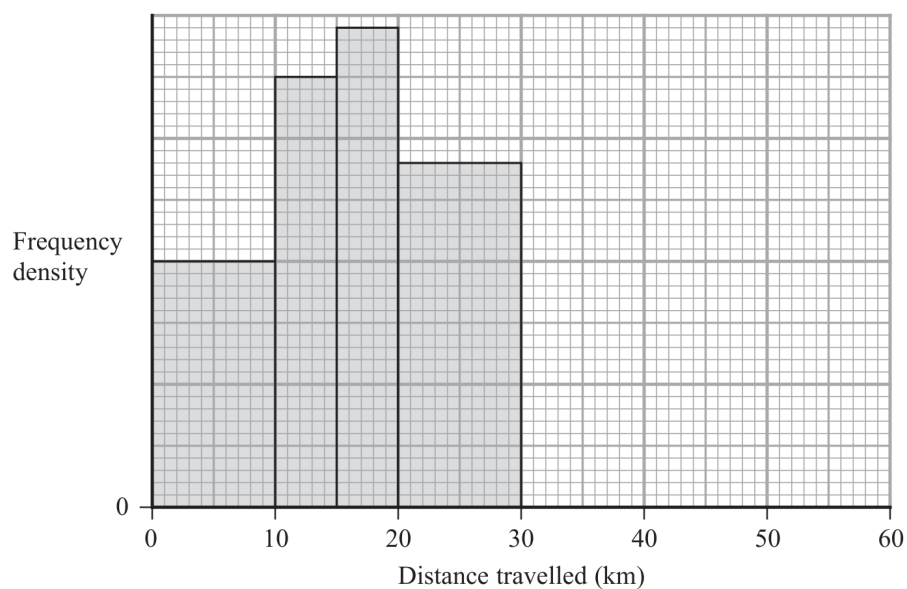
Q21(a) Annotated histogram reading

Density $\times$ class width gives frequency; split 50–80 at 60 and 75Target estimate: $10 \times 2.9 + 15 \times 3.2 + 5 \times 2.0 = 87$ Check: frequency = density $\times$ width

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- 18 The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

Distance (d km)	Frequency
$0 \leq d < 10$	40
$10 \leq d < 15$	
$15 \leq d < 20$	
$20 \leq d < 30$	
$30 \leq d < 60$	30



Using the information in the table and in the histogram,

- (a) complete the table,

(2)

- (b) complete the histogram.

(1)

One of the people working in the store is chosen at random.

- (c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

(2)

Worked solution

June 2021 | Question 18 | 2 marks

Family: Construct and complete histograms • Variant: Complete a table from a partial histogram

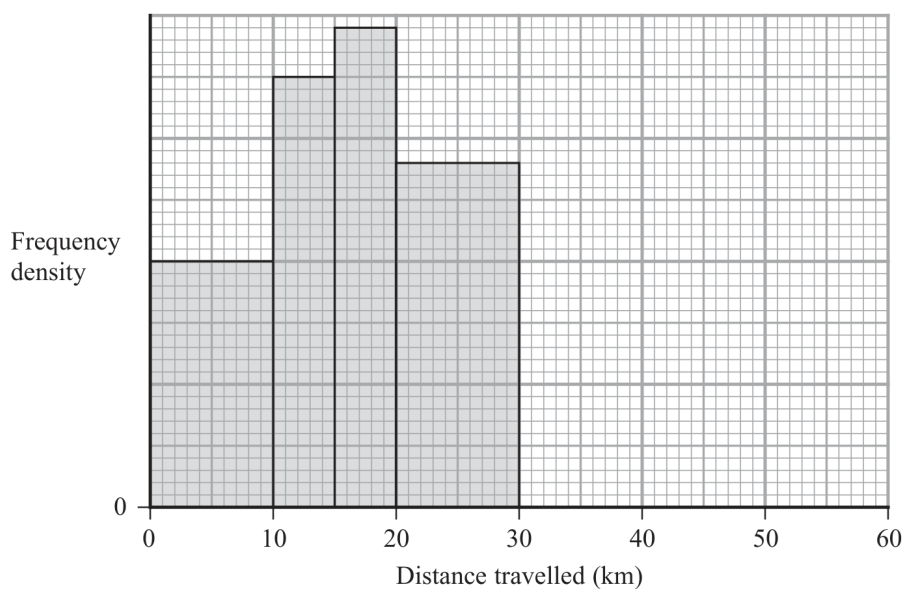
Method / working

1. Use frequency density = frequency / class width to complete the missing table values 35 and 39.
2. Read the remaining frequency from the histogram scale: 56.

Final answer: 35, 39, 56

- 18 The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

Distance (d km)	Frequency
$0 \leq d < 10$	40
$10 \leq d < 15$	
$15 \leq d < 20$	
$20 \leq d < 30$	
$30 \leq d < 60$	30



Using the information in the table and in the histogram,

- (a) complete the table,

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One of the people working in the store is chosen at random.

- (c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

(2)

Worked solution

June 2021 | Question 18 | 1 marks

Family: Construct and complete histograms • Variant: Complete missing histogram bars from mixed information

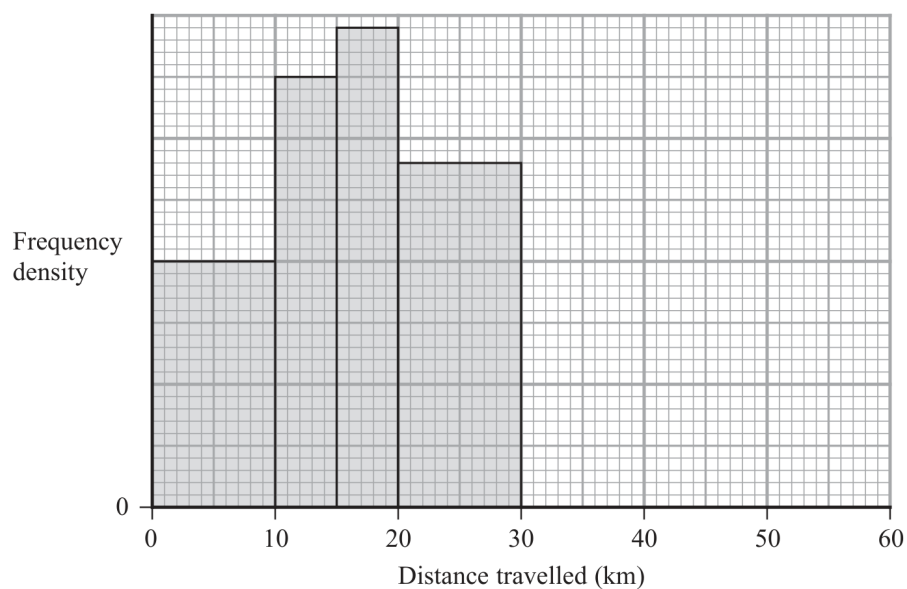
Method / working

1. Complete the 30-60 class with the bar height 1 on the frequency-density scale.

Final answer: Height 1 for $30 \leq d < 60$

- 18 The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

Distance (d km)	Frequency
$0 \leq d < 10$	40
$10 \leq d < 15$	
$15 \leq d < 20$	
$20 \leq d < 30$	
$30 \leq d < 60$	30



Using the information in the table and in the histogram,

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(2)

Worked solution

June 2021 | Question 18 | 2 marks

Family: Interpret histogram areas • Variant: Estimate a histogram proportion, fraction or percentage

Method / working

1. Estimated count above 25 km = $0.5 \times 56 + 30 = 58$ (the 30-60 class contributes 30; do not add the already-used table frequencies again).

2. Total count = 200, so probability = $58/200 = 0.29$.

Final answer: 0.29

4MA1-2H Q16

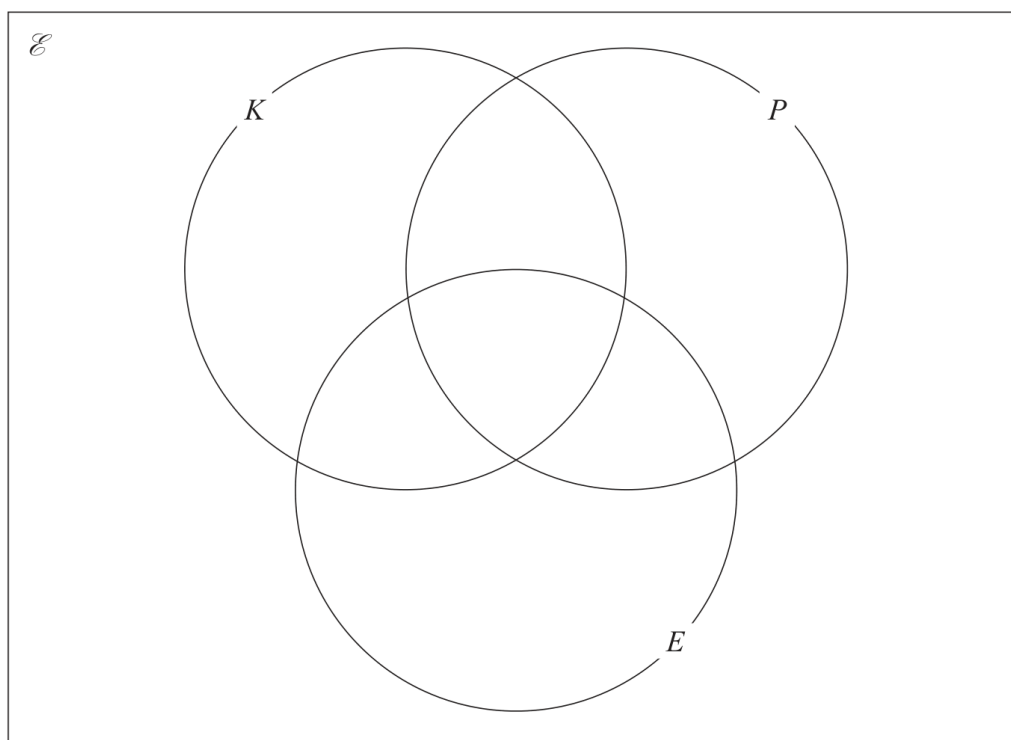
7 marks

- 16 60 art students were asked if they would like to attend workshops for knitting (K), for photography (P) or for embroidery (E)

Of these students

- 9 chose knitting, photography and embroidery
- 17 chose knitting and photography
- 16 chose photography and embroidery
- 20 chose knitting and embroidery
- 28 chose photography
- 39 chose embroidery
- 2 chose none of the workshops

- (a) Using this information, complete the Venn diagram to show the numbers of students in each subset.



(3)

4MA1-2H Q16

7 marks

One of the students is chosen at random.

Given that this student chose photography,

(b) find the probability that this student also chose knitting.

.....
(2)

(c) Find $n(P \cap K')$

.....
(1)

(d) Find $n([P \cup E] \cap K)$

.....
(1)

4MA1-2H Q16

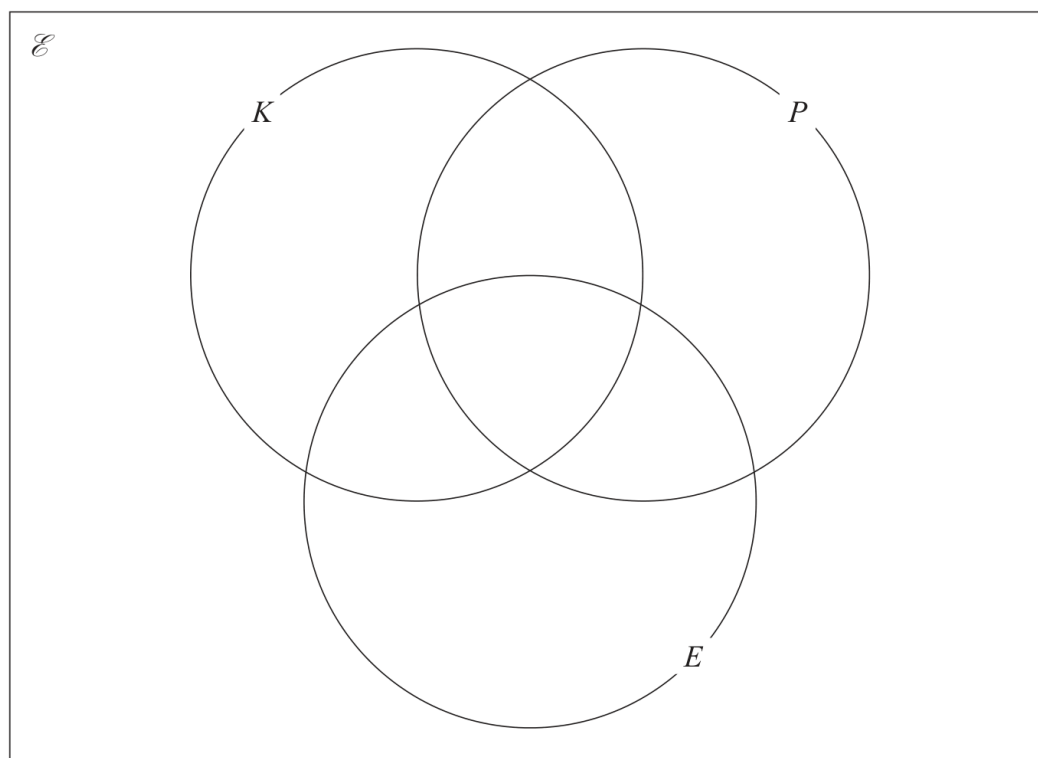
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(3)

4MA1-2H Q16

7 marks

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Given that this student chose photography,

(b) find the probability that this student also chose knitting.

(2)

(c) Find $n(P \cap K')$

(1)

(d) Find $n([P \cup E] \cap K)$

(1)

Three-set Venn diagram

7 marks

Regions

$$KPE = 9, KP = 8, PE = 7, KE = 11, P = 4, E = 12, K = 7, \text{ outside} = 2.$$

Conditional probability

$$P(K | P) = 17/28.$$

Counts

$$n(P \cap K') = 4 + 7 = 11; \quad n((P \cup E) \cap K) = 8 + 11 + 9 = 28.$$

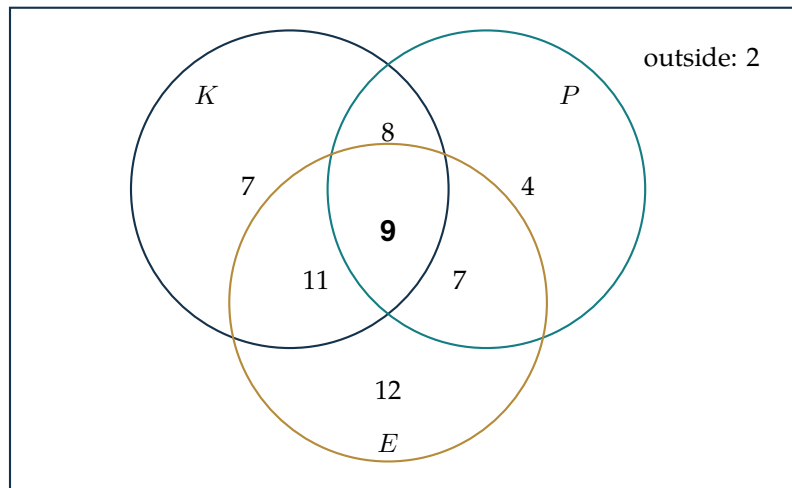
Final answer

$$\frac{17}{28}; \quad 11; \quad 28$$

Three-set Venn diagram

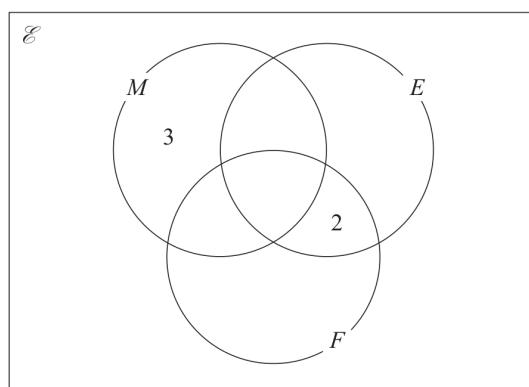
7 marks

Completed solution diagram: Completed three-set Venn diagram



- 16 There are 32 students in a class.
 In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).
 25 students passed the test in Maths.
 20 students passed the test in English.
 14 students passed the test in French.
 18 students passed the tests in Maths and English.
 11 students passed the tests in Maths and French.
 4 students failed all three tests.
 x students passed all three tests.

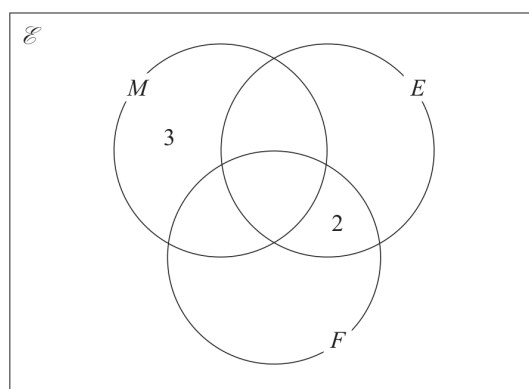
The incomplete Venn diagram gives some more information about the results of the 32 students.



- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

$$x = \dots\dots\dots (2)$$

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

$$\dots\dots\dots (1)$$

Worked solution

November 2021 | Question 16 | 2 marks

Family: Represent and read events with Venn diagrams • Variant: Solve unknown Venn regions using a total

Method / working

1. Use the Maths circle.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $M \text{ only} = 3$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $M \cap E \text{ only} = 18 - x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $M \cap F \text{ only} = 11 - x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. The total in $\setminus(M)$ is $\setminus(25)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $3 + (18 - x) + (11 - x) + x = 25$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $32 - x = 25$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $x = 7$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. So:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $M \cap E \text{ only} = 11$, $M \cap F \text{ only} = 4$, $E \cap F \text{ only} = 2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. $E \text{ only} = 20 - 11 - 2 - 7 = 0$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $F \text{ only} = 14 - 4 - 2 - 7 = 1$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. For a student who passed Maths, failed French means $\setminus(M)$ but not $\setminus(F)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. $\frac{3+11}{25} = \frac{14}{25}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

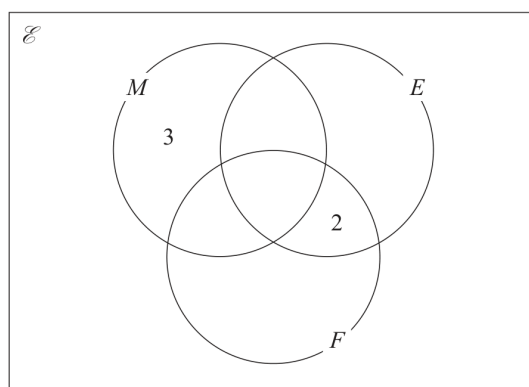
15. Answers: $\setminus(x=7)$, and $\setminus(\frac{14}{25})$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $x=7$, and $\frac{14}{25}$.

- 16** There are 32 students in a class.
 In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).
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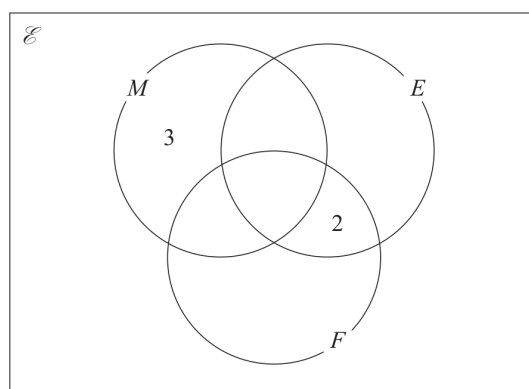
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- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

$x = \dots\dots\dots$
 (2)

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

$\dots\dots\dots$
 (1)

Worked solution

November 2021 | Question 16 | 2 marks

Family: Represent and read events with Venn diagrams • Variant: Complete a three-set Venn diagram

Method / working

1. Use the Maths circle.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $M \text{ only} = 3$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $M \cap E \text{ only} = 18 - x$

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Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. The total in $\setminus(M)$ is $\setminus(25)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $3 + (18 - x) + (11 - x) + x = 25$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $32 - x = 25$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $x = 7$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. So:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

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Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

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14. $\frac{3+11}{25} = \frac{14}{25}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

15. Answers: $\setminus(x=7)$, and $\setminus(\frac{14}{25})$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $x=7$, and $\frac{14}{25}$.

4MA1-1H Q21

3 marks

21 A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

.....
(Total for Question 21 is 3 marks)

WORKING SPACE

4MA1-1H Q21

3 marks

21 A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

.....
(Total for Question 21 is 3 marks)

Exactly two counters of the same colour**3 marks****Use the complement**

$$P(\text{exactly two}) = 1 - P(\text{all same}) - P(\text{all different}).$$

All same

$$P(\text{all same}) = \frac{9 \cdot 8 \cdot 7 + 7 \cdot 6 \cdot 5 + 4 \cdot 3 \cdot 2}{20 \cdot 19 \cdot 18} = \frac{738}{6840}.$$

All different

$$P(\text{all different}) = \frac{6(9 \cdot 7 \cdot 4)}{20 \cdot 19 \cdot 18} = \frac{1512}{6840}.$$

Finish

$$1 - \frac{738 + 1512}{6840} = \frac{4590}{6840} = \frac{51}{76}.$$

Final answer

$$\frac{51}{76}$$

Dependency check Sampling is without replacement; the factor 6 counts the orders of three different colours.

4MA1-2H Q20

6 marks

20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots

12 grow carrots and potatoes and tomatoes

18 grow carrots and potatoes

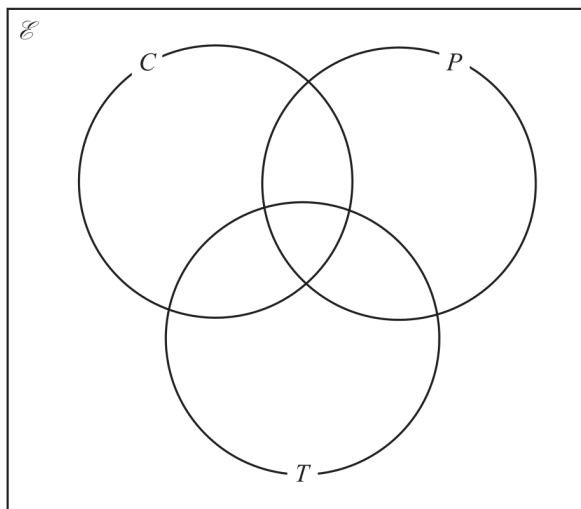
27 grow carrots and tomatoes

32 grow potatoes and tomatoes

29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

(2)

WORKING SPACE

4MA1-2H Q20

6 marks

20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots

12 grow carrots and potatoes and tomatoes

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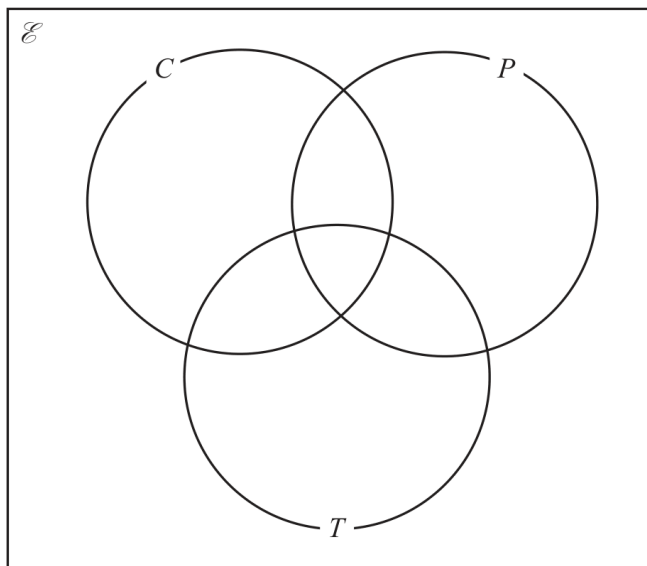
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One of the gardeners who grows carrots is chosen at random.

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(2)

Three-set Venn diagram and conditional probability

6 marks

Intersections

$$C \cap P \cap T = 12, CP \text{ only} = 6, CT \text{ only} = 15, PT \text{ only} = 20.$$

Single-set regions

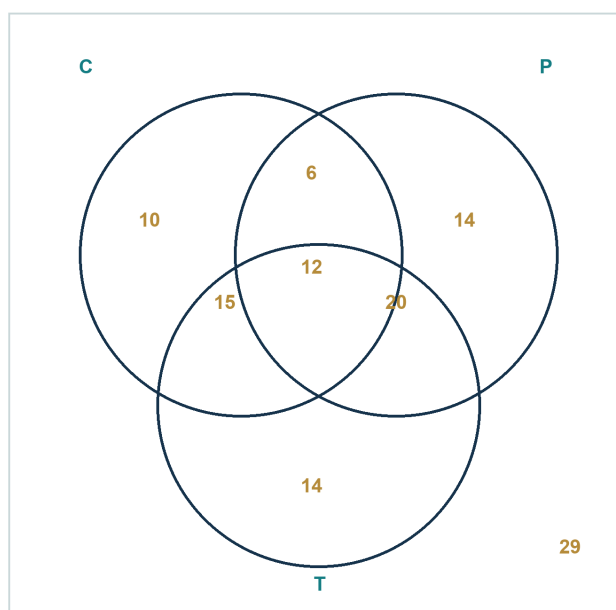
$$C \text{ only} = 43 - (6 + 15 + 12) = 10. \text{ If } P \text{ only} = T \text{ only} = k, 10 + 6 + 15 + 12 + 20 + 2k = 120 - 29, \text{ so } k = 14.$$

Part (b)

$$n(T' \cap C) = 10 + 6 = 16.$$

Part (c)

$$P(P | C) = \frac{n(P \cap C)}{n(C)} = \frac{18}{43}.$$



Solved reference diagram

Final answer

Regions (10, 14, 14, 6, 15, 20, 12; outside 29), 16, $\frac{18}{43}$

Dependency check Parts (b) and (c) read the completed diagram; the diagram appears once.

4MA1-1H Q16

4 marks

16 There are 20 sweets in a box.

15 of the sweets are red

5 of the sweets are yellow

Fred takes at random 3 sweets from the box.

Work out the probability that Fred takes at least one sweet of each colour from the box.

WORKING SPACE

4MA1-1H Q16

4 marks

16 There are 20 sweets in a box.

15 of the sweets are red

5 of the sweets are yellow

Fred takes at random 3 sweets from the box.

Work out the probability that Fred takes at least one sweet of each colour from the box.

At least one of each colour

4 marks

Complement

$$1 - \frac{15}{20} \frac{14}{19} \frac{13}{18} - \frac{5}{20} \frac{4}{19} \frac{3}{18} = \frac{45}{76}.$$

Final answer

$$\frac{45}{76}$$

4MA1-1H Q20 demand

4 marks

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag.
He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

.....

WORKING SPACE

4MA1-1H Q20 demand

4 marks

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The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

.....

Probability of mean 15 cents

4 marks

Demand – Probability of mean 15 cents

Find each coin probability

$$P(10) = \sqrt{49/121} = 7/11, \quad P(20) = 4/11.$$

Use the two orders

$$P(10, 20 \text{ or } 20, 10) = 2 \times \frac{7}{11} \times \frac{4}{11} = \frac{56}{121}.$$

Final answer

56/121

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

8 marks

16 100 farmers are asked if they have goats (G), sheep (S) or chickens (C) on their farms.

Of these farmers

31 have sheep

53 have chickens

6 have goats, sheep and chickens

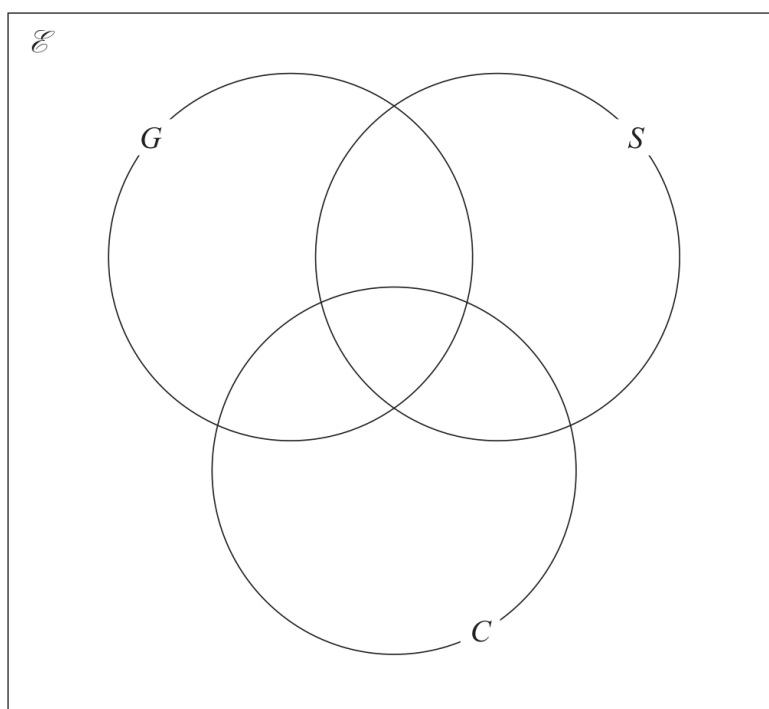
11 have sheep and goats

17 have sheep and chickens

18 have goats and chickens

20 do not have any goats, sheep or chickens

(a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.



(3)

(b) Find

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

8 marks

(i) $n(G)$

(1)

(ii) $n([G \cup S]')$

(1)

(iii) $n(G' \cap C)$

(1)

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

(2)

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

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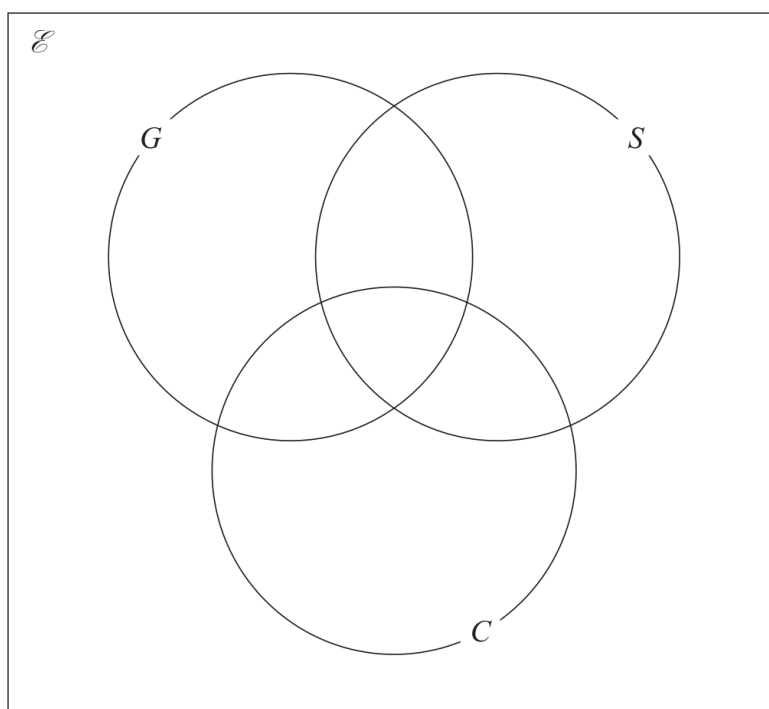
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(3)

(b) Find

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

8 marks

(i) $n(G)$

.....
(1)

(ii) $n([G \cup S]')$

.....
(1)

(iii) $n(G' \cap C)$

.....
(1)

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

.....
(2)

Chapter: Probability Diagrams - Venn & Tree Diagrams; Combined & Conditional Probability

8 marks

Q16(a) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Complete a three-set Venn diagram

Method

Put 6 in the triple intersection; then GS-only = $11 - 6 = 5$, SC-only = $17 - 6 = 11$, and GC-only = $18 - 6 = 12$.

Why: The pair totals include the triple intersection.

Method

S-only = $31 - (5 + 11 + 6) = 9$ and C-only = $53 - (11 + 12 + 6) = 24$.

Why: Subtract every already-filled sheep or chicken region.

Method

There are 80 farmers inside the three sets, so G-only = $80 - (9 + 24 + 5 + 11 + 12 + 6) = 13$; put 20 outside.

Why: Use the universal total and the 20 with none.

Final answer

Venn regions: G only 13; S only 9; C only 24; GS only 5; GC only 12; SC only 11; GSC 6; outside 20

Q16(b)(i) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Method

$n(G) = 13 + 5 + 12 + 6 = 36$.

Why: Add every region inside G.

Final answer

$n(G)$: 36

Q16(b)(ii) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Method

$(G \cup S)'$ contains C-only and the outside region, so $n((G \cup S)') = 24 + 20 = 44$.

Why: The complement excludes every point in G or S.

Final answer

count: 44

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Method

G' intersection C contains C -only and SC -only, so $n(G' \text{ intersection } C) = 24 + 11 = 35$.

Why: Include chicken regions that are outside G .

Final answer

count: 35

Q16(c) – Chapter: Combined & Conditional Probability

Family: Use conditional probability and restricted sample spaces • Variant: Calculate a direct given-that probability

Method

The chosen farmer comes from the 53 farmers with chickens.

Why: This is the conditional sample space.

Method

Of these, 18 also have goats, so the probability is $18/53$.

Why: Use the G -and- C count over the C count.

Final answer

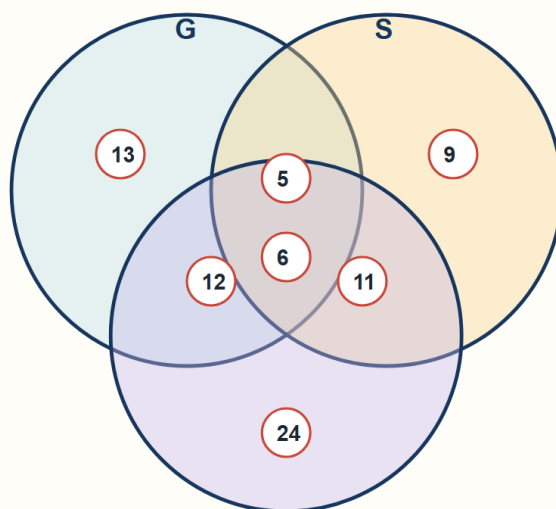
probability: $18/53$

Chapter: Probability Diagrams - Venn & Tree Diagrams; Combined & Conditional Probability

8 marks

Q16(a) Completed three-set Venn diagram

Place every region, including the outside count



outside

20

Inside all three = 6

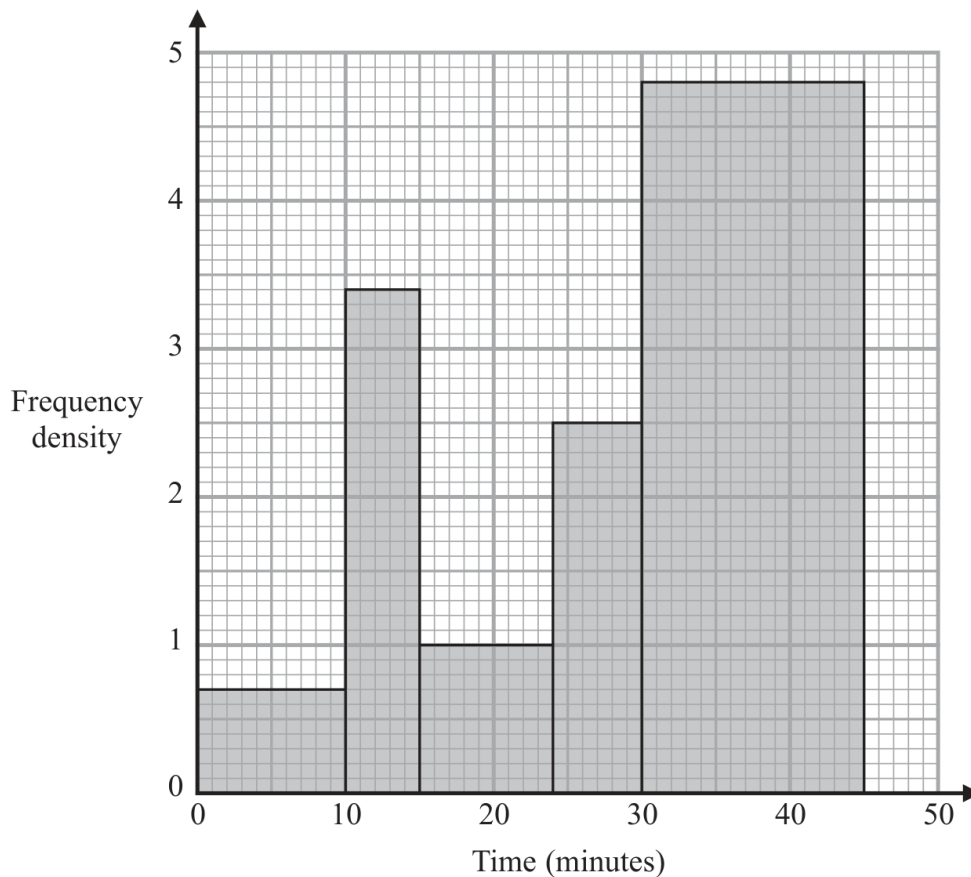
Inside at least one = 80

Total farmers = 100

$80 + 20 = 100$ ✓

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- 18 The histogram gives information about the times, in minutes, that some customers spent in a supermarket.



One of the customers is selected at random.

Given that this customer had spent more than 30 minutes in the supermarket,

- (b) find the probability that this customer spent more than 36 minutes in the supermarket.

(2)

Worked solution

January 2020 | Question 18 | 2 marks

Family: Use conditional probability and restricted sample spaces • Variant: Use a histogram or table as a conditional sample space

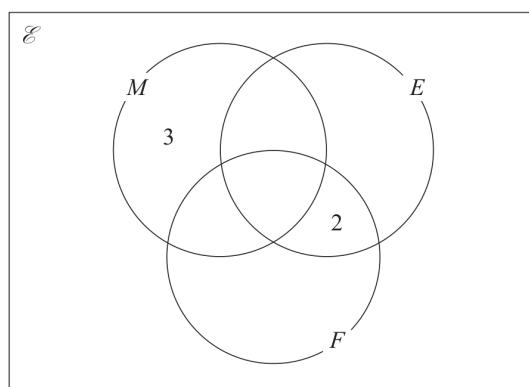
Method / working

1. Given more than 30 minutes, use the 30-45 class.
2. The part above 36 is 9 of the 15 minutes, so probability= $\frac{9}{15}=\frac{3}{5}$.

Final answer: $\frac{3}{5}$

- 16 There are 32 students in a class.
 In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).
 25 students passed the test in Maths.
 20 students passed the test in English.
 14 students passed the test in French.
 18 students passed the tests in Maths and English.
 11 students passed the tests in Maths and French.
 4 students failed all three tests.
 x students passed all three tests.

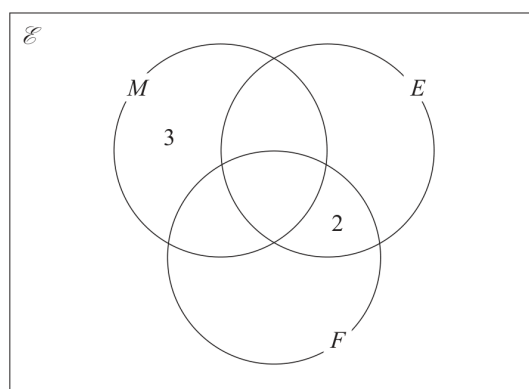
The incomplete Venn diagram gives some more information about the results of the 32 students.



- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

$x = \dots\dots\dots$
 (2)

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

$\dots\dots\dots$
 (1)

Worked solution

November 2021 | Question 16 | 1 marks

Family: Use conditional probability and restricted sample spaces • Variant: Calculate a direct given-that probability

Method / working

1. Use the Maths circle.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $M \text{ only} = 3$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $M \cap E \text{ only} = 18 - x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $M \cap F \text{ only} = 11 - x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. The total in $\setminus(M)$ is $\setminus(25)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $3 + (18 - x) + (11 - x) + x = 25$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $32 - x = 25$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $x = 7$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. So:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $M \cap E \text{ only} = 11$, $\square M \cap F \text{ only} = 4$, $\square E \cap F \text{ only} = 2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. $E \text{ only} = 20 - 11 - 2 - 7 = 0$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $F \text{ only} = 14 - 4 - 2 - 7 = 1$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. For a student who passed Maths, failed French means $\setminus(M)$ but not $\setminus(F)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. $\frac{3+11}{25} = \frac{14}{25}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

15. Answers: $\setminus(x=7)$, and $\setminus(\frac{14}{25})$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $x=7$, and $\frac{14}{25}$.

16 A box contains 15 counters.

There are 4 red counters, 5 green counters and the rest are yellow counters.

Niklas takes at random a counter from the box and writes down the colour of his counter.
He then puts the counter back into the box.

Sasha then takes at random a counter from the box and writes down the colour of her counter.

Work out the probability that the counters taken by Niklas and Sasha both have the same colour.

.....

Worked solution

November 2021 | Question 16 | 3 marks

Family: Combine set counts with sequential probability • Variant: Compare with-replacement and without-replacement models

Method / working

1. There are $\binom{4}{1}$ red, $\binom{5}{1}$ green and $\binom{6}{1}$ yellow counters.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. The first counter is replaced, so the probabilities are the same for Niklas and Sasha.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $P(\text{same colour}) =$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $\left(\frac{4}{15}\right)^2 +$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. $\left(\frac{5}{15}\right)^2 +$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $\left(\frac{6}{15}\right)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $= \frac{16+25+36}{225}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $= \frac{77}{225}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. Answer: $\frac{77}{225}$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $\frac{77}{225}$.

20 A bag contains X counters.

There are only red counters and blue counters in the bag.

There are 4 more blue counters than red counters in the bag.

Finty takes at random 2 counters from the bag.

The probability that Finty takes 2 blue counters from the bag is $\frac{3}{8}$

Work out the value of X .

Show clear algebraic working.

(Total for Question 20 is 5 marks)

Answer reference | May-Nov 2020 | Question 20

Family: Combine selections without replacement • Variant: Recover an unknown population from no-replacement probability

20	$\left(\frac{X+4}{2}\right) \frac{X}{X} (= \frac{X+4}{2X}) \text{ or } \left(\frac{X+4}{2}\right) - 1 (= \frac{X+2}{2X-2})$	eg, where b = number of blue counters $\frac{b}{2b-4} \text{ or } \frac{b-1}{2b-5}$	eg, where r = number of red counters $\frac{r+4}{2r+4} \text{ or } \frac{r+3}{2r+3}$		M1	for making a correct start by finding the probability of the first counter being blue for their method
	eg $\frac{X+4}{2X} \times \frac{X+2}{2X-2}$	eg $\frac{b}{2b-4} \times \frac{b-1}{2b-5}$	eg $\frac{r+4}{2r+4} \times \frac{r+3}{2r+3}$		M1	oe correct calculation for 2 blue (using one variable)
	eg $8(X^2 + 6X + 8) = 3(4X^2 - 4X)$	eg $8b(b-1) = 3(2b-4)(2b-5)$	eg $8(r+4)(r+3) = 3(2r+4)(2r+3)$		M1	dep for a correct equation with no algebraic fractions eg could have $X^2 + 6X + 8 = \frac{3}{8}(4X^2 - 4X)$
	Eg $4X^2 - 60X - 64 (= 0)$ or $X^2 - 15X - 16 (= 0)$ oe	eg $4b^2 - 46b + 60 (= 0)$ or $2b^2 - 23b + 30 (= 0)$ oe	eg $4r^2 - 14r - 60 (= 0)$ or $2r^2 - 7r - 30 (= 0)$ oe		M1	for rearranging their equation to a correct 3 term quadratic
				16	5	A1 cao dep on M4
Total 5 marks						

Worked solution

May-Nov 2020 | Question 20 | 5 marks

Family: Combine selections without replacement • Variant: Recover an unknown population from no-replacement probability

Official final mark-scheme pages are embedded immediately after this question.